AMCD-Net: An Effective Attention-Aided Multilevel Cloud Detection Network for Optical Satellite Imagery

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***Abstract*— Cloud detection is a prerequisite for optical remote sensing applications due to the ubiquitous cloud coverage and negative effect of cloud occlusions. However, it is very challenging because of the heterogeneity of clouds and diversity of underlying surfaces, especially for thin clouds with illegible shapes and dispersed distribution in addition to high transparency, which is a common bottleneck for most methods. To tackle these problems, this article proposes an innovative attention-aided multilevel cloud detection network (AMCD-Net) for optical satellite imagery. Specifically, AMCD-Net takes full account of the variability and complexity of clouds and integrates multilevel features and different attentions within a deep convolutional U-Net framework, thus enabling more accurate cloud identification in complex scenarios. On the one hand, a multilevel asymmetric convolutional (MAC) module embedded encoder is established to learn discriminative representations for clouds with various shapes and integrate the complementarity of multilevel features to improve model robustness, with a regional attention-based decoder constructed to more accurately recover complicated cloud distribution, which effectively balances the integrity and details of clouds. On the other hand, a deformable convolution-based geometry enhancement attention (GEA) is designed to refine information transmission between the encoder and the decoder, with a joint loss of binary cross-entropy (BCE) and structural similarity index (SSIM) constructed to simultaneously focus on category and morphology discriminant excavation, which are favorable for fine-grained cloud prediction. The effectiveness of AMCD-Net was verified on two well-known datasets, i.e., the 38-Cloud dataset and the SPARCS dataset, and the results demonstrate that it outperformed the other state-of-the-art deep networks.**

***Index Terms*— Cloud detection, deep convolutional U-Net, geometry enhancement attention (GEA), joint loss function, multilevel asymmetric convolutional (MAC) module, regional attention.**

I. Introduction

OPTICAL remote sensing images serve as basic data and play an irreplaceable role in various industrial applications, such as agriculture, forestry, environment monitoring, land resource survey, and military defense [1], [2], [3], [4], [5], [6], [7], [8]. According to the statistics of the International Satellite Cloud Climate Program (ISCCP), cloud coverage is commonly found over 66% of the Earth's surface [9], extremely reducing the radiation quality of images and disrupting ground object information extraction, which downgrades the application depth and level of optical remote sensing data to a great extent. Therefore, cloud detection has become an essential preprocessing step for optical satellite imagery in practical applications. However, cloud identification is extremely difficult due to the large variability of clouds and the interference from highly similar underlying surfaces, such as snow/ice, desert, and rocks, especially for thin clouds with high transparency and proximity to clear-sky areas.

Nowadays, numerous cloud detection methods have been developed, which mainly involve single-scene-based and multiscene-based models. Comparatively speaking, multiscene-based models generally yield more accurate cloud masks by introducing additional temporal information for cloud analysis but with limited applications due to an inaccessible data demand for cloud-free reference estimation [10], [11], [12], [13], [14]. Thus, we focus on the single-scene-based models. These models can be further classified into three categories: 1) threshold-; 2) machine learning-; and 3) deep learning-based methods.

The threshold-based methods are the earliest cloud detection models, which distinguish clouds from clear-sky areas with a set of tailored thresholds derived from the high-reflectivity and low-temperature attributes of clouds. Through a long research cycle, a large number of approaches have been proposed for various optical satellites [15]. For example, the automatic cloud cover assessment (ACCA) for Landsat 7 [16], Clouds and the Earth's Radiant Energy System (CERES) for moderate-resolution imaging spectroradiometer (MODIS) [17], and function of mask (Fmask) with enhanced versions for Landsat 4/5/7/8 [18], [19], [20], [21] and Sentinel-2 [22] are typical representatives. These methods mask clouds via a series of threshold tests based on visible and infrared bands. In addition, Sun et al. [23] put forward a universal dynamic threshold cloud detection algorithm for MODIS and Landsat 8 images. Zhai et al. [24] proposed a unified cloud/shadow detection model for multispectral/hyperspectral images based on spectral indices and spatial matching, which achieves competitive performance under eight scenarios acquired by different sensors. These methods have broad

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applications because of simplicity but with limited adaptability to complicated scenarios composed of thin clouds, broken clouds, and clouds over highly similar surfaces, where reasonable thresholds are difficult to determine.

As computer techniques matured, machine learning was naturally introduced to improve cloud detection performance. These approaches infer clouds by learning discriminative differences between clouds and background in the spectral and spatial feature space [25]. For instance, Li et al. [26] detected clouds and cloud shadows for Gaofen-1 (GF-1) imagery by integrating threshold segmentation and guided filter based on multifeatures. By training a support vector machine (SVM) classifier to distinguish clouds and nonclouds, promising results were obtained [27], [28], [29]. In addition, clustering analysis [30], Bayesian model [31], random forest [32], [33], and neural network [34] were also utilized for cloud detection. However, these handcrafted feature-based shallow models fail to learn the intrinsic properties of clouds and display limited robustness to the large variations, which leads to limited detection accuracy.

In recent years, deep learning has shown great advantages to feature extraction and provides a new technical framework for cloud detection. By learning robust high-level semantic features through hierarchical representations, deep learning models significantly advance the development of cloud detection [35], [36], [37]. In specific, Chai et al. [38] employed a deep convolutional neural network (CNN) to establish an AdaptSegNet model for Landsat 8 cloud/shadow detection. Sun et al. [39] combined minimum component transformation with CNN and obtained higher accuracy. Jeppesen et al. [40] proposed a remote sensing network (RS-Net) within a U-Net framework and achieved excellent performance for Landsat 8 imagery. In addition, Zhan et al. [41] utilized a full CNN (FCN) to identify clouds and snow, which delivered accurate results for GF-1 imagery. Zhang et al. [42] further improved cloud detection performance for GF-1 imagery by constructing up and down blocks based on wavelet transform and integrating multiscale spatial attention. Li et al. [43] put forward a weakly supervised cloud detection network that was effective for high-resolution imagery, like GF-1 and Ziyuan-3 (ZY-3) images. Zhao et al. [44] addressed the challenges in cloud boundaries and confused areas by introducing a boundary prediction branch. Hu et al. [45] developed a multibranch convolutional attention network by combining convolutions with transformers, which effectively enhances feature extraction and segmentation capabilities. Zhang et al. [46] proposed a cloud and cloud shadow refinement segmentation network with ResNet-18 as the backbone, which enables more accurate detection of smaller clouds. Furthermore, various cross-sensor models were developed for multisensor cloud detection based on domain adaptation and transformer schemes [47], [48]. Moreover, an increasing number of cloud detection networks have been proposed to further improve identification accuracy and efficiency [49], [50], [51].

Although deep learning networks have greatly advanced cloud detection performance, due to the complexity of clouds and underlying surfaces, challenges remain unresolved in current cloud detection approaches. Firstly, the spectral characteristics and geometric structure of clouds vary greatly, in addition to a complicated distribution, which creates large difficulties for cloud identification and boundary determination. Secondly, the diversity of underlying surfaces and strong interference from similar background objects with high albedo, such as snow/ice, desert/rocks, and artificial surfaces, further increase the challenges. Thirdly, thin clouds have illegible shapes, dispersed distribution, and high transparency, with visually weak features and high proximity to noncloud pixels, which are very difficult to distinguish from the background. As a result, the existing methods still have some deficiencies in handling cloud boundaries and thin clouds, which downgrades the detection accuracy to a certain extent.

In view of these, a novel attention-aided multilevel cloud detection network (AMCD-Net) is proposed for optical satellite imagery in this article. It adopts a deep convolutional U-Net as the skeleton to learn abstract and intrinsic feature expressions for various clouds and takes full advantage of different level features in cloud description and indication to enhance model discriminability and different attentions to promote correct recovery of the complex distribution of clouds, thus boosting more accurate cloud prediction in various scenarios. In general, the main contributions are fourfold.

1) By fully considering the variability of cloud property and irregularity of cloud morphology, an effective multilevel asymmetric convolutional (MAC) module is constructed in the encoder to deeply mine discriminative feature representations for various clouds. Unlike traditional multilevel feature fusion modules, the MAC module has a multipath cross-layer information transmission structure to make full use of initial features from the original data, immediate features from the early layers, and deeply processed features from the previous layer to integrate the complementarity of multilevel and multiscale feature expressions to improve model discriminability and robustness, with stride convolutions used to make up information loss caused by pooling operations. Furthermore, it takes advantage of asymmetric convolutions in irregular target excavation to learn expressive and direction-aware features of clouds. Accordingly, it has a stronger recognition capability and can cope better with thin clouds and cloud boundaries to yield more accurate cloud masks.
2) Considering the complex and imbalanced distribution of clouds, an efficacy regional-attention-based decoder is developed to improve cloud prediction performance. Different from traditional simple feature decoding fashions, it fully considers regional differences and adaptively adjusts channel importance of the extracted features based on region to pay differentiated attention to various regions, which is favorable to fine-grained cloud prediction. In addition, to overcome spatial detail loss caused by traditional linear interpolations, transpose convolutions are used for upsampling to more precisely recover the spatial structure of clouds.
3) Given the high correlation between encoding and decoding and complementarity of different level features, a novel deformable convolution-based geometry

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enhancement attention (GEA) is put forward for skip connections between the encoder and the decoder to learn favorable information from low-level encoding features to aid cloud prediction. It flexibly mines important structure and boundary information of clouds with deformable convolution to generate informative spatial attention to emphasize salient regions while suppressing irrelevant regions, which further integrates multilevel features and contributes to finer morphology prediction and boundary localization.

4) To balance the importance of category and morphology in cloud prediction, a joint loss function consisting of the binary cross-entropy (BCE) loss and the structural similarity index (SSIM) loss is designed instead of traditional individual category loss in most cloud detection networks to guide the network to simultaneously focus on category and morphology discriminant excavation in the training process, which effectively improves the prediction performance and brings clear gains.

The rest of this article is arranged as follows. Section II reviews the related cloud detection networks. Section III details the proposed AMCD-Net. Section IV presents the experimental results, with a detailed performance analysis given in Section V. Section VI summarizes this article and points out the next efforts.

II. Related Work

Given the powerful feature extraction ability of deep learning, various deep networks have been developed for cloud detection and attained remarkable achievements, which greatly improve identification accuracy and facilitate more broad applications of optical remote sensing. These models fall into two categories according to the working mechanism and network architecture, i.e., classification- and prediction-based methods.

Classification-based methods treat the cloud as a specific class and commonly adopt the linear stacking architecture to hierarchically extract discriminative features of clouds and feed them into a classifier to yield a class map with cloud and noncloud labels. Among them, CNNs are the most common framework for cloud classification. For example, Shi et al. [52] put forward a superpixel CNN classification architecture with four convolutional layers and two fully connected layers and trained it for binary classification, which obtained a good effect for Landsat imagery. In addition, Chen et al. [53] developed a multiple CNN composed of three different patch-based branches for cloud classification, with an adaptive simple linear iterative clustering (A-SLIC) algorithm to generate high-quality superpixels, which performed well for ZY-3 and GF-1 imagery. Wang et al. [54] proposed an object-based two-branch CNN to mine multiscale features, which distinguished clouds and snow well for Sentinel-2 imagery. Furthermore, Luotamo et al. [55] constructed a dual CNN with a branch for inner cloud identification and another branch for cloud edge refinement, which learns features from a large spatial extent along with classification. Moreover, Segal-Rozenhaimer et al. [47] employed a multilayer perceptron neural network (MLP-NN) with four hidden layers and VGG-16-based CNN with five convolutional blocks to deeply mine spectral–spatial features to classify clouds and cloud-free pixels based on a soft-max classifier, which performed well for multimodal applications, such as worldview-2 and Sentinel-2 imagery [47].

Prediction-based methods commonly adopt the U-Net architecture and include two parts, i.e., encoder and decoder, which are responsible for cloud discriminative feature extraction and spatial distribution prediction, respectively, to yield a probability map with values ranging from 0 to 1, thus indicating the cloud likelihood for each pixel. Then, a threshold is required to generate the final cloud mask. Comparatively speaking, this kind of method is more suitable for large-scene applications, with more abundant examples. For instance, Li et al. [56] proposed a multiscale convolutional feature fusion (MSCFF) model for high-resolution image cloud and cloud shadow detection, which utilizes a CBRR block including convolution, batch normalization (BN), rectified linear unit (ReLu), and residual unit to build a deep encoder and decoder, with a multiscale module for feature fusion. Dev et al. [57] developed a lightweight CloudSegNet by employing simple convolutional layers, pooling layers, and upsampling layers to construct the encoder and the decoder for efficient cloud segmentation. In addition, Zhang et al. [50] proposed a lightweight U-Net by using depth-separable convolution to reduce the computational cost of the model, which obtained good results for Landsat-8 imagery.

Furthermore, Mohajerani and Saeedi [49] developed a Cloud-Net based on specially designed convolutional blocks to construct a deep architecture to fully exploit the semantic features of clouds, which yielded promising results for Landsat-8 imagery. Based on this, Kanu et al. [58] put forward a CloudX-Net that utilizes separable convolution and atrous spatial pyramid pooling (ASPP) to optimize the network and further improve cloud detection performance. Jeppesen et al. [40] proposed an advanced RS-Net based on the U-Net architecture, establishing a deep encoder and decoder with shortcut connections between them for information transmission, which delivered competitive performance for Landsat-8 imagery. Moreover, Liu et al. [51] developed a DC-Net by introducing deformable convolution blocks into the encoder to more accurately model cloud morphologies. Li et al. [59] proposed a robust global context-dense block U-Net (GCDB-UNet) by embedding a global context mining block into U-Net to improve thin cloud identification ability. Besides, Chen et al. developed a multiscale attention feature aggregation network (MAFA-Net) [60] and a multiscale strip feature attention network (MSFA-Net) [61], with ResNet-18 as the backbone. The former used multiscale bar pool attention to extract multiscale contextual and channel information, while the latter introduced boundary detail feature awareness and cross-layer self-attention feature fusion modules to focus on boundaries, which obtained state-of-the-art cloud detection performance. Li et al. [62] proposed a new residual dual U-shape network (RD-UNet) by combining two U-Nets and using residual connections to enhance information flow, which yielded more accurate results but had a large number of parameters and high hardware requirements.

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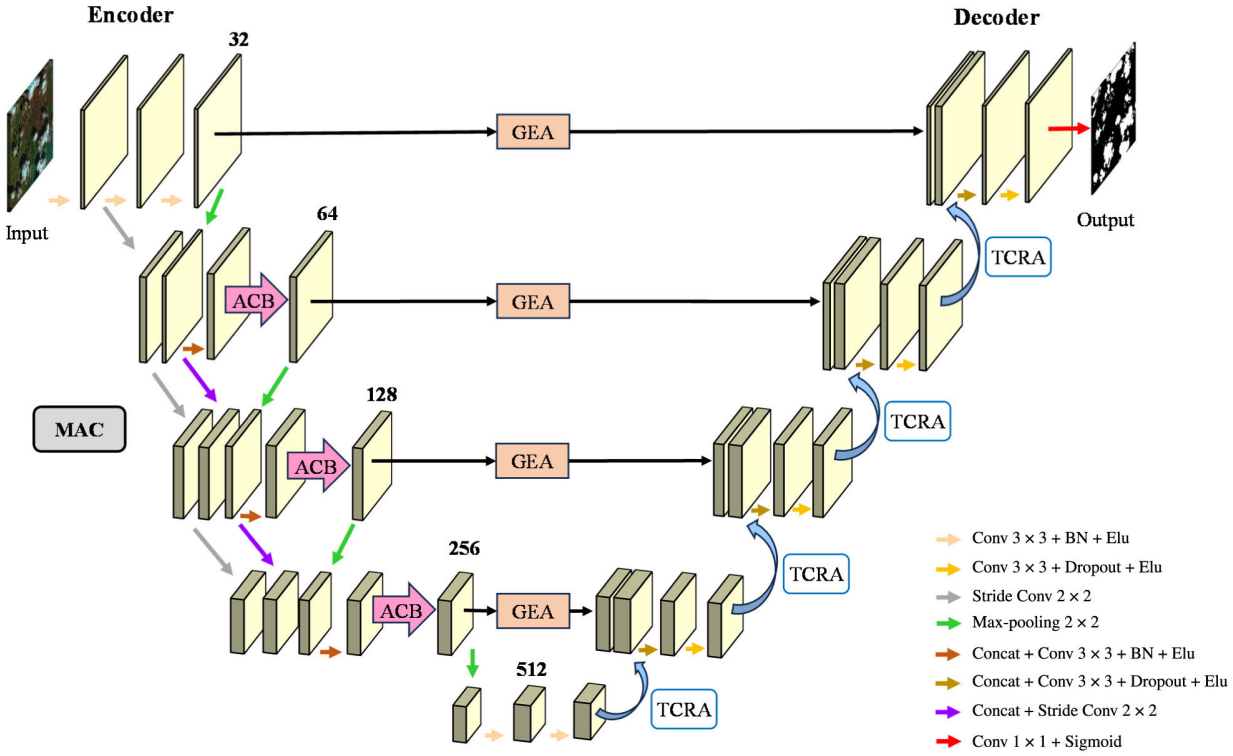


Fig. 1. Overview of the whole framework of AMCD-Net.

In summary, a large number of advanced deep networks have been proposed for cloud detection, which greatly promotes the development of this field. However, these methods still have difficulties in dealing with cloud boundaries and thin clouds, which limits their accuracy and applicability to a certain degree.

III. Attention-Aided Multilevel Cloud Detection Network

In this section, an effective AMCD-Net is proposed for optical satellite imagery to promote accurate cloud detection in complex scenarios. Within a U-Net framework, AMCD-Net learns expressive features for various clouds by combining conventional convolutions with asymmetric convolutions and integrates the advantages of multilevel feature representations and different attentions to further enhance model discriminability to yield more accurate cloud masks.

A. Overall Architecture

The whole framework of AMCD-Net is depicted in Fig. 1. As observed, it adopts a deep convolutional U-Net architecture, with an encoder for semantic feature extraction and a decoder for spatial distribution prediction. The encoder consists of five layers, with the decoder composed of four layers. In addition, in this network, skip connections are constructed between the corresponding encoding and decoding layers for information transmission. With an image patch $\mathbf{I} \in \mathbb{R}^{h\times w\times c}$ as the input, where h and w denote the height and width, with c being the number of channels, a pixelwise prediction probability map $\mathbf{O} \in \mathbb{R}^{h\times w\times 1}$ can be produced.

In the encoder, a standard 3×3 convolution is firstly implied for the input data to extract shallow features. Subsequently, an MAC module consisting of four downsampling layers follows to learn abstract and invariant feature expressions for various clouds with irregular shapes layer-by-layer and integrate complementary information of multilevel features. The first layer consists of two conventional convolutions to extract primary features, with the other three layers composed of one conventional convolution for multilevel feature integration and one asymmetric convolutional block (ACB) for direction-aware representation learning. A max-pooling operation is deployed at the end of each layer to downsample the features derived by ACB. In addition, multipath cross-layer feature transmissions are deployed based on stride convolutions to deduce more expressive fused features. Then, through the last convolutional layer composed of two conventional convolutions, the final encoding feature can be obtained.

In the decoder, four upsampling layers are deployed, with each layer consisting of a transposed convolution with a regional attention (TCRA) block for feature upsampling and regional differentiated cloud recovery and two traditional convolutions for feature integration and refinement. Herein, transposed convolution is used to replace traditional linear interpolation for upsampling, which can effectively reduce detail loss and more finely recover cloud distributions. In addition, a GEA is designed for skip connections to extract favorable information from the low-level encoding features and suppress useless information to generate more informative features to aid decoding. Then, two conventional convolutions follow to integrate and refine the features derived by TCRA and GEA to further improve prediction performance. Through

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progressive decoding, a feature map with the same resolution as the original image patch can be obtained. Finally, a cloud probability map is yielded through one 1×1 convolution and a sigmoid function. With a proper threshold, the final cloud mask can be generated.

B. MAC Module Embedded Encoder

As is well known that clouds generally have large variability and complex morphologies, which reduces the efficacy of traditional convolution learning, therefore, it is essential to introduce a new feature extraction fashion to more effectively mine expressive features for irregular clouds. In addition, different level features have clear differences in object description, with low-level features paying more attention to visual characteristics in terms of color, shape, and boundary, while high-level features focus on abstract semantic representations. Hence, how to fuse the complementarity of multilevel features is crucial to improve the cloud detection effect. At this point, an effective MAC module is proposed, which introduces asymmetric convolutions to overcome the defects of conventional convolutions in irregular target description and makes full use of multilevel feature representations through multipath cross-layer information transmission to enhance the discriminability and robustness of the model.

To learn expressive features for clouds with diverse shapes, asymmetric convolutions are used to extract local contextual information along different directions. Inspired by AC-Net [63], three main directions are considered for feature extraction to form an ACB, as shown in Fig. 2. The primary feature is simultaneously fed into three parallel branches with different kernels to focus on discriminability excavation in different directions, with one square kernel of 3×3 for main body extraction and two kinds of asymmetric kernels of 1×3 and 3×1 for shape detail extraction along horizontal and vertical directions. Then, the features derived by each branch are integrated through weighted fusion to induce the final refined feature, as formulated in the following equation:

$$\mathbf{F}_{out} = w_1\delta(\mathbb{N}(\mathbf{F}_{in} * K_{3\times3})) \oplus w_2\delta(\mathbb{N}(\mathbf{F}_{in} * K_{1\times3})) \oplus w_3\delta(\mathbb{N}(\mathbf{F}_{in} * K_{3\times1})) \quad (1)$$

where $\mathbf{F}_{in}$ and $\mathbf{F}_{out}$ denote the input and output features; $K_{3\times3}$, $K_{1\times3}$, and $K_{3\times1}$ stand for the square, horizontal, and vertical kernels, respectively; $*$ and $\oplus$ represent convolution and elementwise addition operations; and $\mathbb{N}(\cdot)$ and $\delta(\cdot)$ refer to BN and Elu activation, with w_1, w_2, and w_3 being the weights for features from the three branches, which are set as 1, 1/2, and 1/2 here. In this way, more discriminative direction-aware feature representations can be obtained, which is favorable to fine-grained cloud prediction.

With ACB as the core, an innovative MAC module with four downsampling layers and three information transmission paths is constructed, as described in Fig. 1. The first path is used for initial feature transmission, which inherits primary properties from the original data. The second path is used for immediate feature transmission, which contains integrated information from the early layers. The third path is used for deeply processed feature transmission, which contains discriminative information derived from the previous layer. Through multipath information transmission and periodic comprehensive learning, complementary information from multilevel features with different scales and receptive fields can be fully fused, which further enriches feature expressiveness and further improves model robustness. In this way, the MAC module well balances the integrity and details of clouds, in addition to improving thin cloud identification and boundary localization abilities.

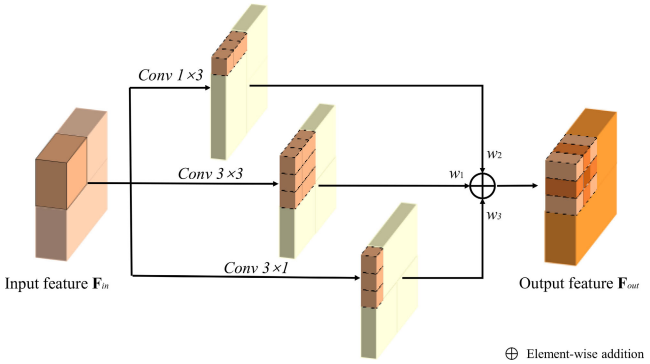


Fig. 2. Diagram of ACB.

Specifically, the first layer of the MAC module is formed by two 3×3 conventional convolutions and one 2×2 max-pooling, with the other layers consisting of a conventional 3×3 convolution, an ACB, and a 2×2 max-pooling. Considering the scale difference caused by max-pooling, 2×2 stride convolutions are employed to process cross-layer features to facilitate information fusion, which effectively reduces feature size and alleviates information loss. The transmitted features are fused by concatenation and used for further discriminative information mining.

With the MAC module as the main body, a deep encoder that consists of five layers can be established. To increase the richness of encoding features, 32, 64, 128, 256, and 512 convolutional kernels are deployed for the five layers, respectively. Herein, 3×3 kernels are used for local content learning, with the padding operation used to keep feature size unchanged. The exponential linear unit (Elu) is used for nonlinear activation, with 2×2 max-pooling and 2×2 stride convolution employed for feature downsampling. After four layers of downsampling, the feature size becomes 1/16 of the original image patch. Then, the feature is fed into the last convolutional layer. With two 3×3 conventional convolutions for further feature refinement, the final encoding feature is generated, which contains rich semantic information and is the basis of the decoder.

C. Regional Attention-Based Decoder

Considering the complex and uneven distribution of clouds, a natural idea is to differentially regard various regions in the cloud prediction process to highlight cloud concentration regions and suppress background regions to more precisely recover the spatial distribution of clouds. In addition, although traditional linear interpolation-based upsampling can restore the resolution of feature maps, it inevitably leads to certain

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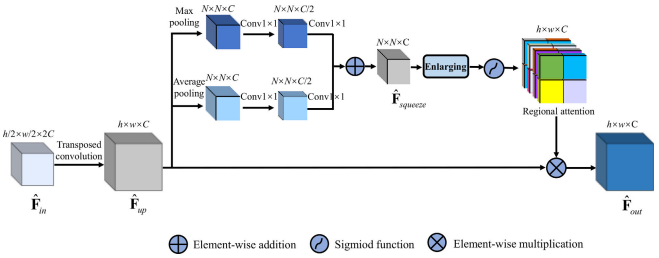
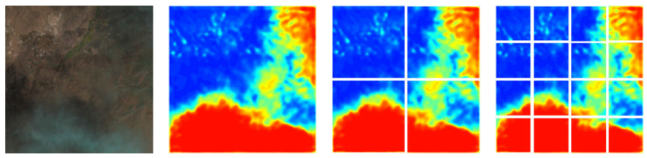


Fig. 3. Diagram of the TCRA module.



(a) (b) (c) (d)

Fig. 4. Uneven distribution of clouds: (a) image patch, (b) no region division, (c) 2×2 region division, and (d) 4×4 region division.

detail loss, which limits the prediction performance to some degree. In this context, an efficiency regional attention-based decoder is developed, which employs transposed convolutions for feature upsampling to better reserve cloud details and fully exploits regional differences to generate more accurate cloud predictions.

Firstly, a TCRA block is constructed in each decoding layer for feature upsampling and regional refinement, as depicted in Fig. 3. It consists of two key parts, i.e., transposed convolution and regional attention. For an input preliminary feature $\hat{\mathbf{F}}_{in} \in \mathbb{R}^{h/2\times w/2\times 2C}$, through a transposed convolution with a kernel size of 3×3 and a step length of 2, the channel number is reduced to half and the spatial size is doubled. As a result, an upsampled feature $\hat{\mathbf{F}}_{up} \in \mathbb{R}^{h\times w\times C}$ is obtained, which is fed into regional attention for refinement. Regional attention is a novel region-based channel of attention [64]. Different from traditional global channel attention, it adopts a region division strategy to avoid global pooling and learns regional differentiated channel importance to refine the input feature. As depicted in Fig. 4, due to the uneven distribution of clouds, global pooling confuses clouds with the background, making global channel attention lose efficacy for emphasizing clouds. Through region division, clouds and background can be separated, with the learned attention effectively enhancing the importance of cloud concentration regions while suppressing background. Therefore, it is conductive to promoting correct prediction of clouds.

Specifically, the feature maps are divided into $N\times N$ regions, with N=2 in the former two layers for low-resolution features and N=4 in the latter two layers for high-resolution features. Subsequently, the maximum and average pooling operations are employed to compress the spatial information for each region, with two 1×1 convolutions used to extract channel importance on the basis of regional features. Then, by fusing the derived features through an elementwise addition, a squeezed feature $\hat{\mathbf{F}}_{squeeze} \in \mathbb{R}^{N\times N\times C}$ can be generated as follows:

$$\hat{\mathbf{F}}_{squeeze} = Conv_1\left(Conv_1\left(P_m\left(\hat{\mathbf{F}}_{up}\right)\right)\right) \oplus Conv_1\left(Conv_1\left(P_A\left(\hat{\mathbf{F}}_{up}\right)\right)\right) \quad (2)$$

where $P_m(\cdot)$ and $P_A(\cdot)$ denote the maximum pooling and average pooling, with $Conv_1(\cdot)$ standing for the 1×1 convolution. Then, through enlarging and sigmoid activation, regional attention $\mathbf{A}_r \in \mathbb{R}^{h\times w\times C}$ is induced, which contains regional differentiated channel importance. Finally, based on $\mathbf{A}_r \in \mathbb{R}^{h\times w\times C}$, a refined feature $\hat{\mathbf{F}}_{out} \in \mathbb{R}^{h\times w\times C}$ can be obtained as follows:

$$\hat{\mathbf{F}}_{out} = S\left(\gamma\left(\hat{\mathbf{F}}_{squeeze}\right)\right) \otimes \hat{\mathbf{F}}_{up} \quad (3)$$

where $S(\cdot)$ and $\gamma(\cdot)$ denote the sigmoid function and element replication-based enlarging operation, with ⊗ representing the elementwise multiplication operation. In this way, the spatial details of clouds can be more accurately restored.

Then, with the TCRA block as the core, a deep decoder consisting of four upsampling layers can be established. For each layer, the upsampled feature derived by the TCRA block is fused with the refined low-level encoding feature from skip connection through concatenation, and then, two conventional 3×3 convolutions follow to integrate and refine the fused feature for the next upsampling. After four rounds of the same operations and one 1×1 convolution together with a sigmoid function, an accurate cloud probability map can be obtained.

D. Skip Connections With GEA

Numerous studies have indicated that skip connections are very beneficial for improving prediction performance by propagating favorable encoding features to aid decoding. However, directly transmitting low-level encoding features to the decoding path leads to relatively higher information redundancy, which interferes with important spatial information excavation and limits the prediction accuracy to some degree. Therefore, by fully considering the complexity of cloud shapes and boundaries, a new deformable convolution-based GEA is developed to extract important spatial structural information from low-level decoding features to promote more accurate restoration of spatial details of clouds.

The diagram of GEA is shown in Fig. 5. With a low-level encoding feature $\mathbf{F}^l$ from the lth layer as the input, GEA generates a refined feature $\tilde{\mathbf{F}}^l$ based on an attention map. Herein, deformable convolution is employed for structural information excavation from $\mathbf{F}^l$ to enhance geometric representations. Different from conventional convolutions with square kernels and fixed reception field, deformable convolution increases offsets for each sampling position, which enables convolutional kernels to deform freely, as described in Fig. 6 [65]. It can effectively reduce the interferences of irrelevant factors contained in the window and improve the orientation perception ability. In this way, the complex cloud shapes and boundaries can be flexibly modeled. For a kernel with M sampling positions, the deformable convolution can be formulated in the following equation:

$$\mathbf{F}_{dc}(q) = \sum_{i=1}^{M} w_i \cdot \mathbf{F}(q + q_i + \Delta q_i) \quad (4)$$

where $\mathbf{F}_{dc}(q)$ and $\mathbf{F}(q)$ represent the features at location q of the output and input feature maps, with w_i, q_i, and Δq_i referring to the weight, preset offset, and offset for the

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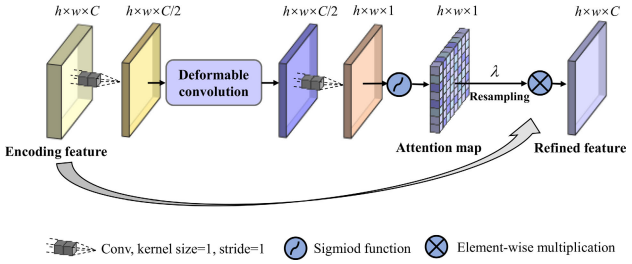


Fig. 5. Diagram of the GEA module.

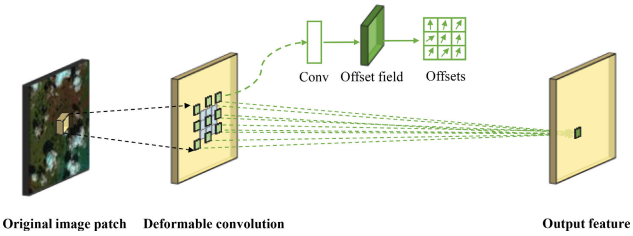


Fig. 6. Illustration of deformable convolution.

ith sampling position, respectively. Here, a 3×3 kernel with nine sampling positions is used for deformable convolution.

To reduce the large computational complexity of deformable convolution, a 1×1 convolution is firstly used to reduce the channel number by half and extract channel correlations, with the result $\boldsymbol{\varphi}^l$ fed into a deformable convolution to mine spatial structural information. Then, with a 1×1 convolution for channel compression and through a BN operation and a sigmoid activation, a spatial selective attention map $\mathbf{A}^l$ can be induced as follows:

$$\mathbf{A}^l = S\left(\mathbb{N}\left[\mathbf{W}_1^T\left(DC\left(\boldsymbol{\varphi}^l\right)\right) + \mathbf{b}_1\right]\right) \tag{5}$$

where $\mathbf{W}_1$ and $\mathbf{b}_1$ denote the parameters of 1×1 convolution, with $DC(\cdot)$ representing deformable convolution. Each element of $\mathbf{A}^l$ can be considered as a spatial weight for the corresponding pixel, which indicates the importance of the current position.

Finally, by multiplying the resampled attention map with the encoding feature $\mathbf{F}^l$, a refined feature $\tilde{\mathbf{F}}^l$ can be obtained, as formulated in the following equation:

$$\tilde{\mathbf{F}}^l = \mathbf{F}^l \otimes \left(\lambda \cdot \mathbf{A}^l\right) \tag{6}$$

where λ denotes a coefficient factor, which is set as 1/2 in practice. In this way, the refined feature only focuses on salient regions of interest and adaptively adjusts the contributions of different spatial positions in the low-level encoding feature, which is conductive to improving cloud prediction accuracy.

E. Joint Loss Function

Given that the quality of cloud prediction results is usually determined by two factors, i.e., category and morphology, it is reasonable to take them into account in the loss function to improve its monitoring and guidance capabilities for network training. Therefore, an effective weighted joint loss function is constructed instead of the traditional individual category loss to better train the proposed AMCD-Net.

Specifically, the joint loss function consists of two items, i.e., BCE loss and SSIM loss. The BCE loss is widely used in various tasks [66], such as classification, segmentation, and change detection, which makes the network focus on category information learning and guides the network to mine more discriminative features to distinguish clouds and background. The SSIM is a common metric for image similarity evaluation, which measures the similarity of two images by comparing their brightness, contrast, and structure [67]. Here, we use it as a morphology loss to guide the network to learn a more similar prediction with the ground truth. By integrating them with proper weights, a joint loss function can be obtained as follows:

$$\ell_J = \alpha \ell_{BCE}\left(L, \hat{L}\right) + (1-\alpha)\ell_{SSIM}\left(L, \hat{L}\right) \tag{7}$$

$$\ell_{BCE}\left(L, \hat{L}\right) = \sum_{i=1}^{h}\sum_{j=1}^{w}\left(L_{ij}\log\left(\hat{L}_{ij}\right) + \left(1-L_{ij}\right)\log\left(1-\hat{L}_{ij}\right)\right) \tag{8}$$

$$\ell_{SSIM}\left(L, \hat{L}\right) = 1 - \frac{\left(2\mu_L\mu_{\hat{L}} + c_1\right)\left(2\sigma_{L\hat{L}} + c_2\right)}{\left(\mu_L^2 + \mu_{\hat{L}}^2 + c_1\right)\left(\sigma_L^2 + \sigma_{\hat{L}}^2 + c_2\right)} \tag{9}$$

where $\hat{L} \in \mathbb{R}^{h\times w}$ and $L \in \mathbb{R}^{h\times w}$ denote the produced cloud mask and ground truth, respectively; ℓ_J, ℓ_{BCE}, and ℓ_{SSIM} represent the joint loss, BCE loss, and SSIM loss; α is the weight for BCE loss, which is set as 0.7 in practice; μ_L, $\mu_{\hat{L}}$ and σ_L, $\sigma_{\hat{L}}$ are expectations and variances of L and $\hat{L}$; $\sigma_{L\hat{L}}$ is their covariance; c_1 and c_2 are two constants to maintain stability. The values in the cloud mask $\hat{L}$ range from 0 to 1, with a larger value indicating a higher probability for the corresponding pixel belonging to the cloud. This optimization problem can be easily solved by the Adam optimizer [68].

In this way, the cloud category and morphology can be simultaneously monitored in the training process, which effectively improves the prediction performance and helps with boundary localization.

IV. Experiment and Analysis

A. Dataset Description

The performance of the proposed AMCD-Net is testified on two well-known datasets, i.e., the 38-Cloud dataset and the SPARCS dataset, which are publicly available datasets with various kinds of clouds and underlying surfaces and widely used for cloud detection performance evaluation.

The 38-Cloud dataset uses Landsat-8 Operational Land Imager (OLI) images as the data source and consists of 38 typical scenes, which is a very challenging dataset for cloud detection [49]. It includes various kinds of clouds, such as thick clouds, thin clouds, and broken clouds, with cloud amounts ranging from 0% to 100%. In addition, the underling surfaces are diverse, including barren, grass/crop, snow/ice, and water. In some scenarios, the background is highly similar to clouds, which greatly increases the difficulty of cloud detection. Following the common strategies [49], 18 images were utilized for training, with 20 images utilized for testing. Each image was cropped into 384×384 patches. As a result, 8400 and 9201 patches were obtained for training and testing.

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Furthermore, 20% of the training set was split for verification, resulting in 6720 patches for training and 1680 patches for verification.

The SPARCS dataset gathered 80 typical scenes of Landsat-8 OLI images, with each scene having a smaller size of 1000×1000 pixels [34]. This dataset has a variety of cloud types, cloud amounts, and underlying surfaces as well. The original annotation includes seven categories, i.e., cloud, cloud shadow, shadow over water, snow/ice, water, land, and flooded. For convenience, the cloud was treated as an individual category, with the rest grouped into a unified category, i.e., background. In addition, the underlying surfaces were placed into similar classes to the first dataset, i.e., barren, grass/crop, snow/ice, and wetland. Similarly, each image was cropped into 384×384 patches, with 540 and 180 patches obtained for training and testing. Then, 20% of the training set was split for verification.

During data preprocessing, all images were normalized, with pixel values ranging from 0 to 1. The size of the input patch was adjusted to 192×192. In addition, data augmentation was implied to enrich the dataset and guarantee the training of various networks, such as horizontal flipping, vertical flipping, rotation, and zooming, which is especially important for the SPARCS dataset with a relatively small size.

B. Experimental Setup

To verify the effectiveness of the proposed AMCD-Net, eight representative and state-of-the-art cloud detection networks were selected as benchmarks, including Cloud-Net [49], CloudX-Net [58], AdaptSegNet [38], RS-Net [40], DC-Net [51], MAFA-Net [60], MSFA-Net [61], and RD-UNet [62]. For fare comparison, all methods share the same dataset partition and augmentation strategies. In addition, for all comparative methods, the parameters were set as recommended.

The proposed AMCD-Net was implemented in Keras (2.4.3) with TensorFlow (2.4.0). In the training process, the learning rate was set as 0.0001, and the batch size was set as 24 and 12, with the epoch number set as 200 and 600 for the 38-Cloud dataset and the SPARCS dataset, respectively. BN was employed for each convolutional layer in the encoder with a momentum of 0.7. In the testing phase, a cloud probability map was obtained for each patch based on the well-trained network. It was binarized with a global threshold of 0.3 for the 38-Cloud dataset and 0.35 for the SPARCS dataset and then resized to 384×384 to yield the final cloud mask.

All experiments were carried out on a server with Intel[1] Xeon[1] w-2245 CPU, 64-GB memory, and NVIDIA GeForce RTX 3090 GPU. Five independent experiments were executed for each method, with average evaluations reported. For a comprehensive analysis of different methods, both qualitative visual maps and quantitative evaluations were given. Herein, five common metrics were utilized in the comparisons, i.e., precision, recall, overall accuracy (OA), F1_score, and Jaccard index, which can be defined as follows:

$$\text{precision} = TP/(TP + FP) \tag{10}$$

$$\text{recall} = TP/(TP + FN) \tag{11}$$

$$\text{OA} = (TP+TN)/(TP + TN + FP + FN) \tag{12}$$

$$\text{F1-score} = (2 \times \text{precision} \times \text{recall}) /(\text{precision} + \text{recall}) \tag{13}$$

$$\text{Jaccard index} = TP/(TP + FN + FP) \tag{14}$$

where TP, TN, FP, and FN denote the number of true positive, true negative, false positive, and false negative pixels in the confusion matrix derived by cloud mask and reference. Precision gives insight into misinspection, while recall reflects missed inspection. OA, F1-score, and Jaccard are three comprehensive metrics, which combine these two aspects. However, OA does not account for the imbalance between the target and background, which cannot accurately indicate the performance in the cases when the background dominates. Overall, these metrics evaluate the result from different perspectives and can well indicate cloud detection performance.

C. Results on the 38-Cloud Dataset

The cloud detection performance of different methods was firstly tested on the 38-Cloud dataset, with quantitative evaluations given in Table I. As illustrated, DC-Net performed badly on this dataset due to its simple structure. The low recall of 74.69% suggested serious cloud omissions. As a result, the F1-score and Jaccard were only 78.83% and 70.27%, which are far from satisfactory. Cloud-Net and AdaptSegNet effectively reduced omissions and significantly outperformed DC-Net, by introducing deep convolutional architectures to mine more discriminative features. By using separable convolutions and ASPP to optimize the network, CloudX-Net further improved the detection performance. RS-Net exhibited a large potential in cloud detection to yield higher precisions, owing to a more powerful encoder and decoder.

In contrast, the recently developed MAFA-Net and MSFA-Net models attained competitive performance on this dataset by introducing multiscale attentions to learn contextual and channel information and multiscale strip feature attentions to enhance boundary mining ability. However, the relatively lower recalls of 82.55% and 82.74% led to unsatisfactory results. By employing dual residual U-Nets to help each other to enhance discriminability excavation, RD-UNet delivered outstanding performance on this dataset to yield the second-best results, with a high F1-score of 87.71% and a high Jaccard of 80.65%.

Comparatively speaking, the proposed AMCD-Net further improved the detection performance, by integrating the advantages of multilevel features and different attentions, to achieve the best result on this dataset with clear gains in most indicators over the other methods. Specifically, it achieved an obviously higher recall of 88.78%, which indicates a significant reduction in the omission rate of clouds, with the suboptimal precision of 91.82% obtained, which effectively ensures the correct detection of clouds. As a result, it achieved the optimal values in comprehensive indicators, with the highest OA of 96.51%, F1-score of 89.23%, and Jaccard of

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TABLE I
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE 38-CLOUD DATASET (%)

Methods	Precision	Recall	OA	F1-score	Jaccard
Cloud-Net	89.88	84.18	95.96	84.24	77.18
CloudX-Net	91.38	84.24	95.55	86.38	78.49
AdaptSegNet	89.54	83.96	96.02	84.80	77.10
RS-Net	90.22	86.92	96.21	87.24	79.87
DC-Net	87.37	74.69	94.95	78.83	70.27
MAFA-Net	89.81	82.55	95.95	84.93	77.22
MSFA-Net	89.00	82.74	95.97	84.23	76.49
RD-UNet	**92.16**	86.75	96.50	87.71	80.65
AMCD-Net	91.82	**88.78**	**96.51**	**89.23**	**82.10**

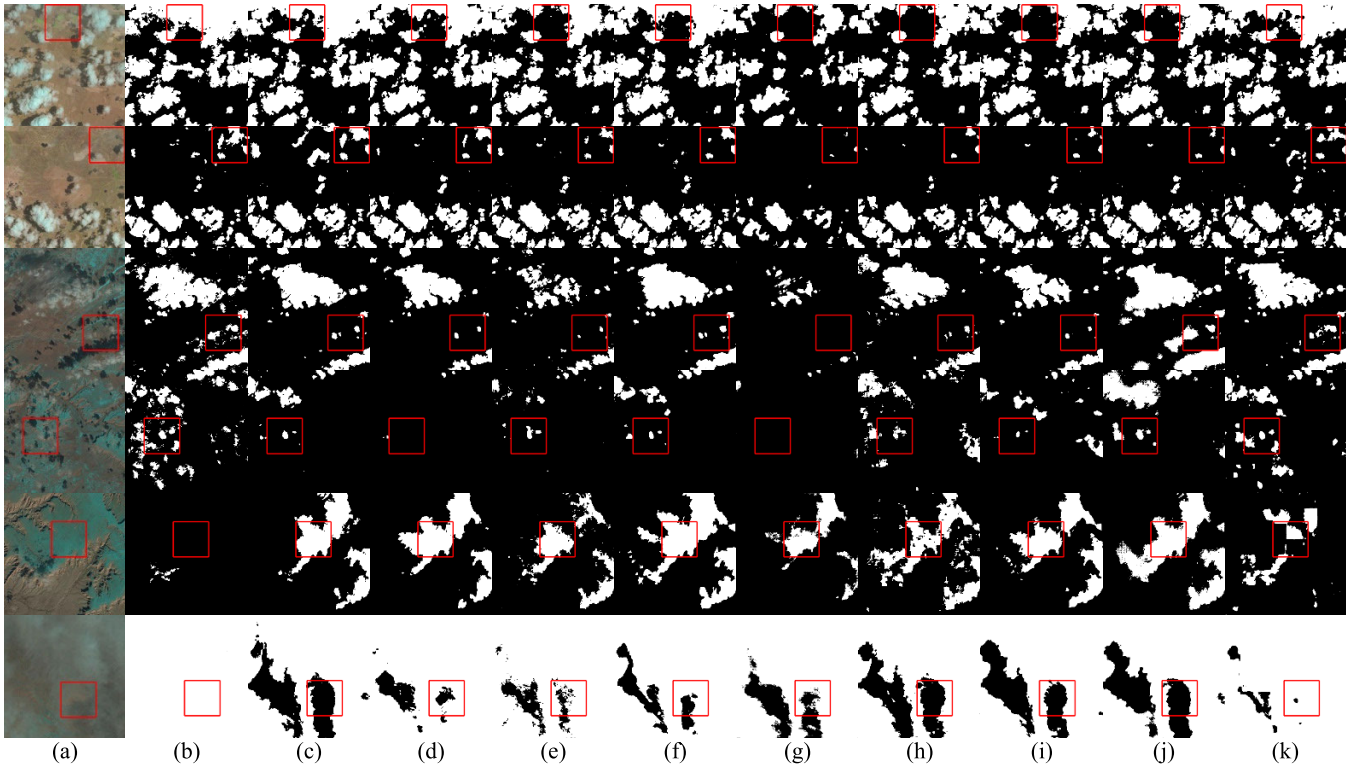


Fig. 7. Cloud masks of different methods for the barren scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

82.10%. Compared with the second-best method, i.e., RD-UNet, AMCD-Net attained obvious improvements of nearly 1.5% in terms of F1-score and Jaccard index. With respect to the other methods, more significant gains were attained, which demonstrates the effectiveness and superiority of the proposed method.

The visual effect of cloud masks obtained by different methods was also compared. For a comprehensive comparison, six representative scenes were selected for each kind of underlying surfaces, with cloud masks provided in Figs. 7–11. As can be seen, the brightness and texture of different underlying surfaces vary greatly, which leads to different challenges for cloud detection. Overall, AMCD-Net achieved the best visual effect on the 38-Cloud dataset to yield cloud masks closest to the references, with good performance for both thick and thin clouds over various underlying surfaces, which further indicates its effectiveness and robustness.

Specifically, for the barren scenes, clouds are difficult to distinguish from highly similar background objects, as shown in Fig. 7. It can be found that most detection methods are effective for thick clouds but with weak identification ability when dealing with thin clouds. For example, the comparative methods more or less missed the thin clouds around thick clouds in the red boxes of the first and second rows, while AMCD-Net effectively alleviated this problem and well detected thick clouds with more accurate details and boundaries. In the third and fourth rows, the benchmark approaches lost efficacy for the visually weak thin clouds in the red boxes, whereas AMCD-Net successfully identified them to yield more accurate cloud masks. Furthermore, in the fifth row, it can be observed that the benchmarks failed to distinguish clouds from the strong interference background, with a large amount of pseudo clouds in the results. Comparatively, AMCD-Net overcame these defects to effectively identify the clouds hidden in the background and reduce false detections, with a more

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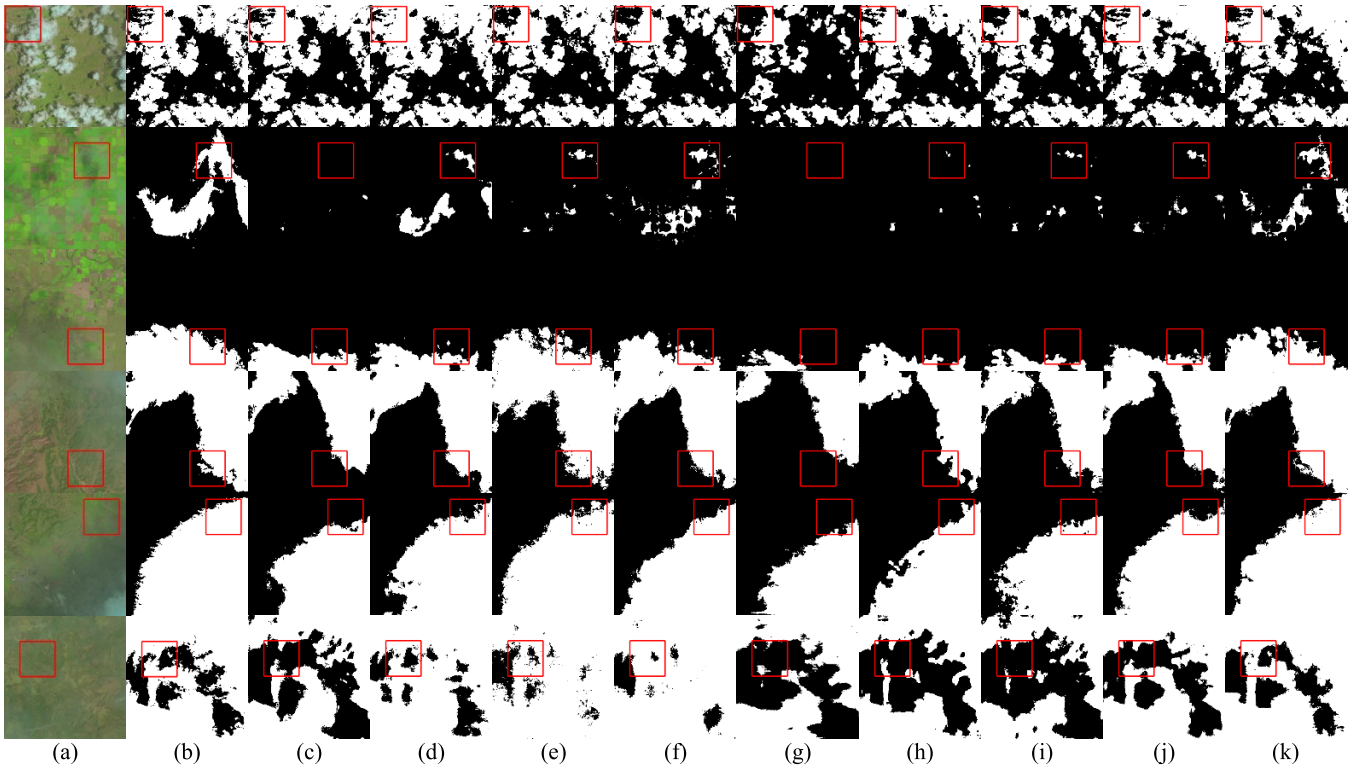


Fig. 8. Cloud masks of different methods for the grass/crop scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

promising result yielded. Moreover, for the large-area thin clouds over bare surfaces as in the sixth row, serious missed inspections appeared in the results of comparative methods, while AMCD-Net significantly refined the cloud masks with fewer omissions.

For the grass/crop scenes in Fig. 8, AMCD-Net outperformed the other methods again. Most methods missed the thin clouds around thick clouds in the red box of the first row, while CloudX-Net, MAFA-Net, and RD-UNet overestimated the thin clouds. By comparison, AMCD-Net effectively identified the thin clouds and well maintained the structure and boundaries of thick clouds. For the visually weak thin clouds as in the second and third rows, the benchmark methods lost efficacy, while AMCD-Net successfully detected most thin clouds to yield more accurate results. For the large-area thin clouds in the fourth, fifth, and six rows, most methods had more or less missed inspections, and CloudX-Net, AdaptSegNet, and RS-Net had serious overestimations, whereas AMCD-Net performed much better to yield more promising cloud masks with finer boundaries.

The snow/ice scenes are most challenging for cloud detection due to the strong interference from snow and ice that share high albedo and low temperature with clouds, as shown in Fig. 9. It can be observed that, in the red boxes of the first and second rows, there was a large quantity of missed inspections or misinspections in the results of comparative methods. In contrast, AMCD-Net overcame these obstacles to yield more accurate cloud masks. For the large-area thin clouds in the rest rows, different degrees of omissions downgraded the performance of benchmarks, especially for CloudX-Net and DC-Net, while AMCD-Net effectively detected most thin clouds to yield refined cloud masks.

For the water scenes in Fig. 10, similar conclusions can be drawn. As illustrated, DC-Net performed poorly for thin clouds and broken clouds, with serious omissions and coarse cloud masks. Cloud-Net, CloudX-Net, and AdaptSegNet performed much better by significantly reducing omissions but losing efficacy for visually weak thin clouds, as observed in the red boxes. MAFA-Net and MSFA-Net delivered competitive performance but still lacked strength for thin clouds. RD-UNet further improved the performance by reducing the omissions of thin clouds. By comparison, AMCD-Net achieved the most accurate cloud masks in all cases, with refined structure and boundaries.

Furthermore, to more intuitively evaluate the performance of the proposed AMCD-Net, two typical whole scenes were selected for each underlying surface, including both the thick cloud scenario and thin cloud scenario, with the corresponding cloud masks given in Fig. 11. As can be observed, AMCD-Net successfully detected both thick clouds and thin clouds over various underlying surfaces. Regardless of macro-integrity and micro-local details, AMCD-Net performed well, which further demonstrates its effectiveness and practicability.

D. Results on the SPARCS Dataset

The cloud detection performance of different methods was further compared on the SPARCS dataset, with quantitative evaluations given in Table II. Similar to the first experiment,

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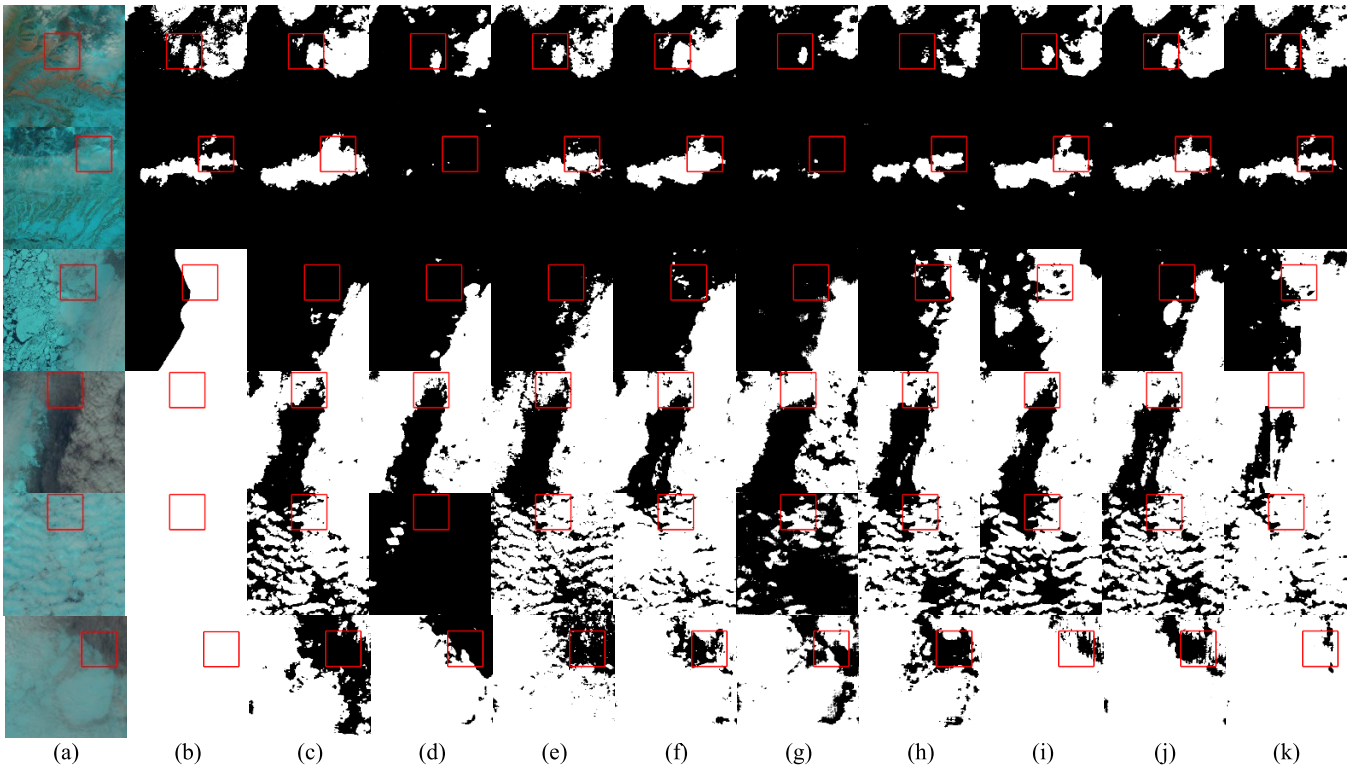


Fig. 9. Cloud masks of different methods for the snow/ice scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

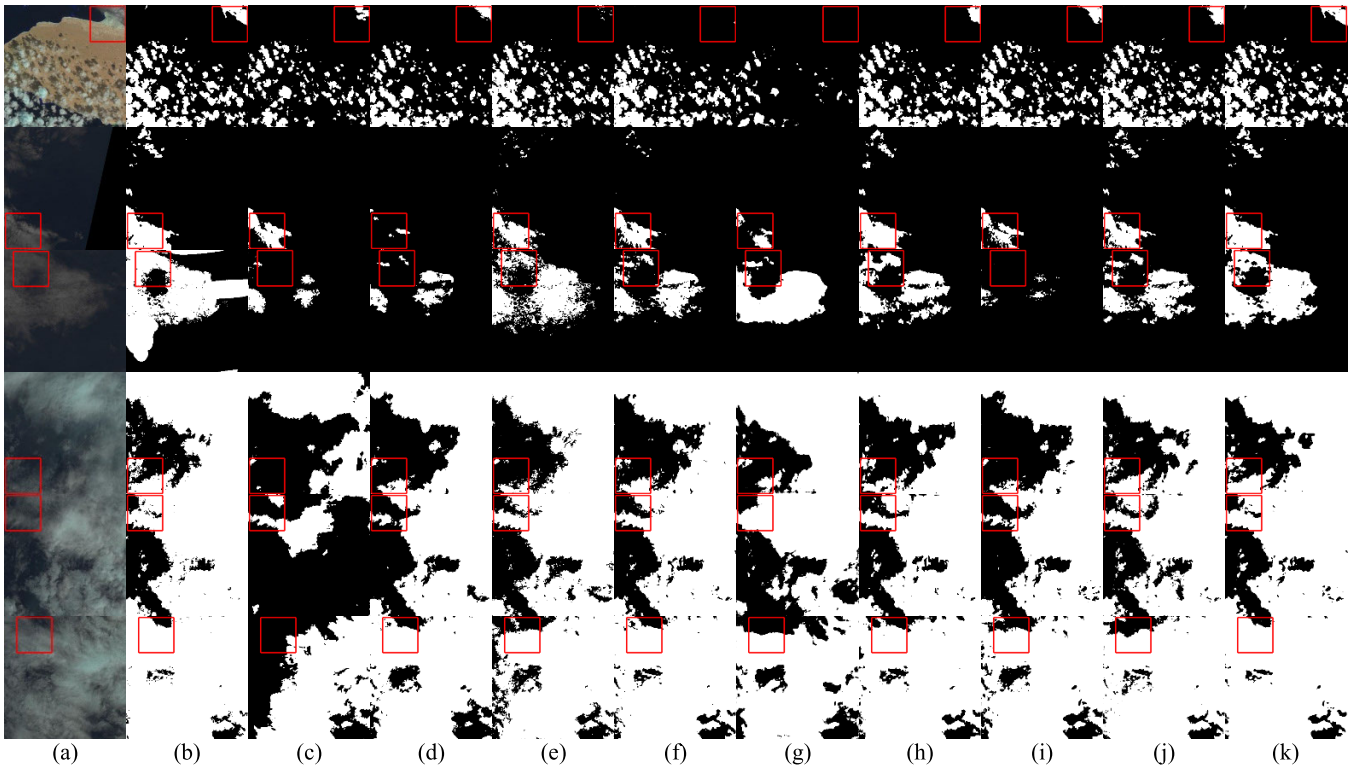


Fig. 10. Cloud masks of different methods for the water scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

DC-Net performed poorly on this dataset, with a low F1-score of 70.38% and Jaccard of 58.13%. Cloud-Net, CloudX-Net, and AdaptSegNet outperformed DC-Net by reducing the omission rates, with much higher F1-scores and Jaccard indices obtained. MAFA-Net, MSFA-Net, and RD-UNet further improved the detection performance to deliver higher

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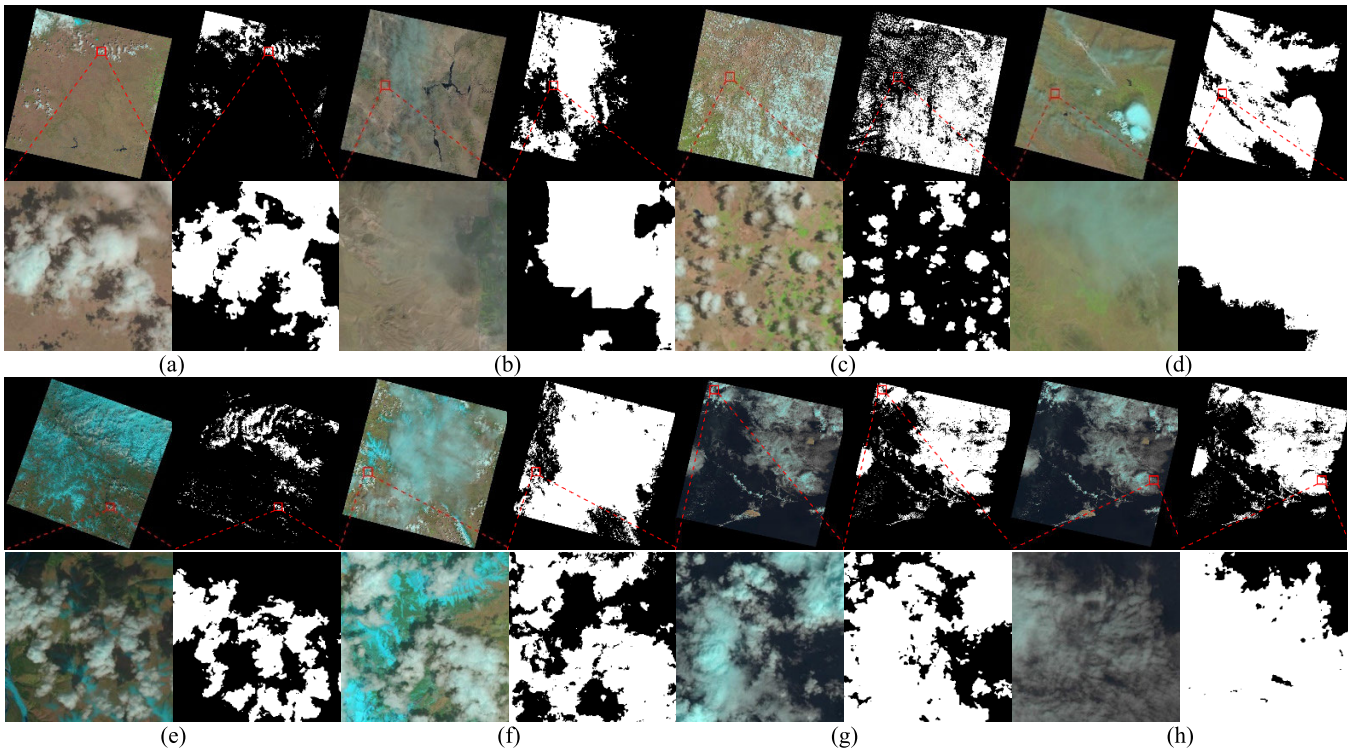


Fig. 11. Typical whole scenes and the corresponding cloud masks obtained by AMCD-Net over various underlying surfaces. (a) Scene ID: LC08_L1TP_032035_20160420_20170223_01, barren. (b) Scene ID: LC08_L1TP_039035_20160320_20170224_01, barren. (c) Scene ID: LC08_L1TP_029044_20160720_20170222_01, grass/crop. (d) Scene ID: LC08_L1TP_034029_20160520_20170223_01, grass/crop. (e) Scene ID: LC08_L1TP_064015_20160420_20170223_01, snow/ice. (f) Scene ID: LC08_L1TP_034033_20160520_20170223_01, snow/ice. (g) Scene ID: LC08_L1TP_003052_20160120_20170405_01, water. (h) Scene ID: LC08_L1TP_003052_20160120_20170405_01, water.

TABLE II
QUANTITATIVE EVALUATIONS OF DIFFERENT METHODS ON THE SPARCS DATASET (%)

Methods	Precision	Recall	OA	F1-score	Jaccard
Cloud-Net	89.61	80.35	91.83	81.04	72.25
CloudX-Net	91.24	79.67	92.28	81.60	73.05
AdaptSegNet	90.32	79.25	91.83	81.37	72.31
RS-Net	91.13	83.41	93.13	84.39	76.46
DC-Net	**97.10**	59.27	88.84	70.38	58.13
MAFA-Net	89.37	82.46	92.37	82.62	74.05
MSFA-Net	90.12	83.96	92.97	84.41	75.79
RD-UNet	90.17	83.10	92.93	83.59	75.54
AMCD-Net	90.75	**86.86**	**93.90**	**87.02**	**79.17**

precisions. RS-Net performed competitively on this dataset to yield the second-best results.

By comparison, AMCD-Net performed much better, as shown by most indicators. Specifically, it achieved a much higher recall of 86.86% and a high-level precision of 90.75%, which suggested that it minimized missed inspections while ensuring correct detection of clouds. Accordingly, AMCD-Net achieved the highest F1-score of 87.02% and Jaccard index of 79.17%, with obvious improvements of nearly 3% compared with the second-best method and more prominent gains over the other methods. Overall, AMCD-Net achieved the best result again on this dataset, which further indicates its effectiveness.

Furthermore, the visual maps of different methods were compared, with cloud masks of four typical scenes for each kind of underlying surfaces shown in Figs. 12–16. Similar conclusions can be drawn as in the first experiment. As illustrated, DC-Net failed to distinguish clouds from the background well in complex scenarios, such as barren and snow/ice, yielding coarse cloud masks with a large number of omissions and false detections. Comparatively, Cloud-Net, CloudX-Net, AdaptSegNet, and RS-Net performed much better for thick clouds and obvious thin clouds, with fewer omissions, but had weak ability for visually weak thin clouds and large-area thin clouds, as shown in the red boxes. MAFA-Net, MSFA-Net, and RD-UNet delivered competitive performance and obtained relatively accurate cloud masks, but they still lacked strength when dealing with thin clouds and cloud boundaries, with unsatisfactory visual effects. In contrast, AMCD-Net performed well for both thick and thin clouds in various cases, yielding more accurate cloud masks that better maintained cloud skeletons and boundaries. Overall,

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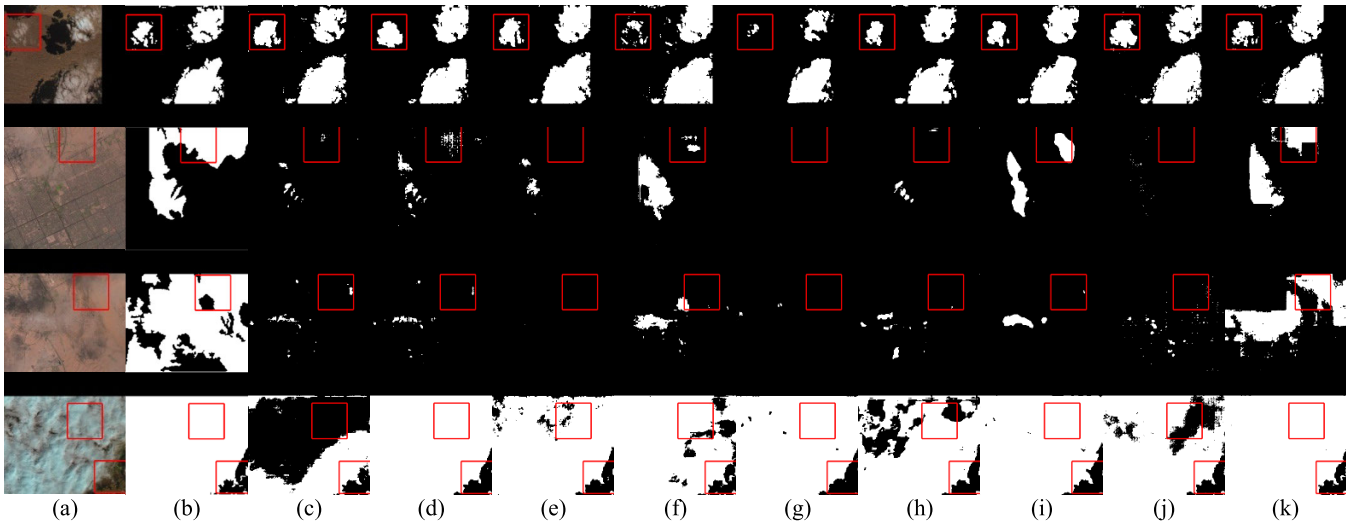


Fig. 12. Cloud masks of different methods for the barren scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

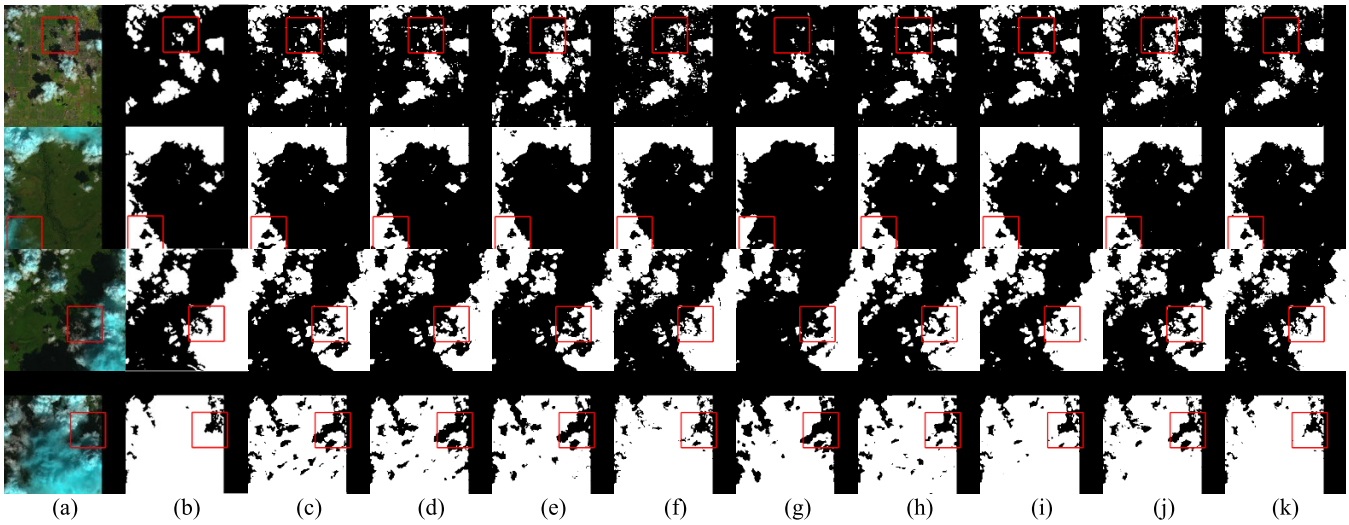


Fig. 13. Cloud masks of different methods for the grass/crop scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

AMCD-Net achieved the best visual effect over various underlying surfaces, with cloud masks closest to the reference.

To further evaluate the visual effect of AMCD-Net, similarly, two typical whole scenes were selected for each underlying surface, with cloud masks given in Fig. 16. As can be observed, AMCD-Net achieved cloud masks of high quality for each scene, which are basically consistent with the cloud distribution in the original images. Whether for thick clouds or thin clouds, it performed well over various underlying surfaces, with refined cloud details and boundaries, as seen in the local magnifications, which demonstrates its robustness.

V. Discussion

A. Performance Analysis in Different Scenarios

Underlying surfaces commonly have significant differences, which lead to different challenges for cloud detection. To further analyze the robustness of different methods and their adaptability to different scenarios, cloud detection performance over each kind of underlying surfaces was separately evaluated on both the 38-Cloud dataset and the SPARCS dataset, with detailed results given in Tables III and IV. As can be observed, AMCD-Net achieved the best performance in all cases on both two datasets, yielding the highest evaluations in most indicators.

Specifically, DC-Net performed poorly in various scenarios, with low-level precisions obtained, due to the serious omissions indicated by the low recalls. Comparatively speaking, Cloud-Net, CloudX-Net, AdaptSegNet, and RS-Net effectively improved the detection performance to yield much higher accuracy. However, these methods still had limited performance in the challenging barren and snow/ice scenarios, especially for the SPARCS dataset, due to the omission of thin clouds in complex background. MAFA-Net, MSFA-Net, and RD-UNet faced similar problems, lacking strength for visually insignificant thin clouds, which limits the detection accuracy to some extent. Benefitting from the strong recognition and structural information mining capabilities brought by

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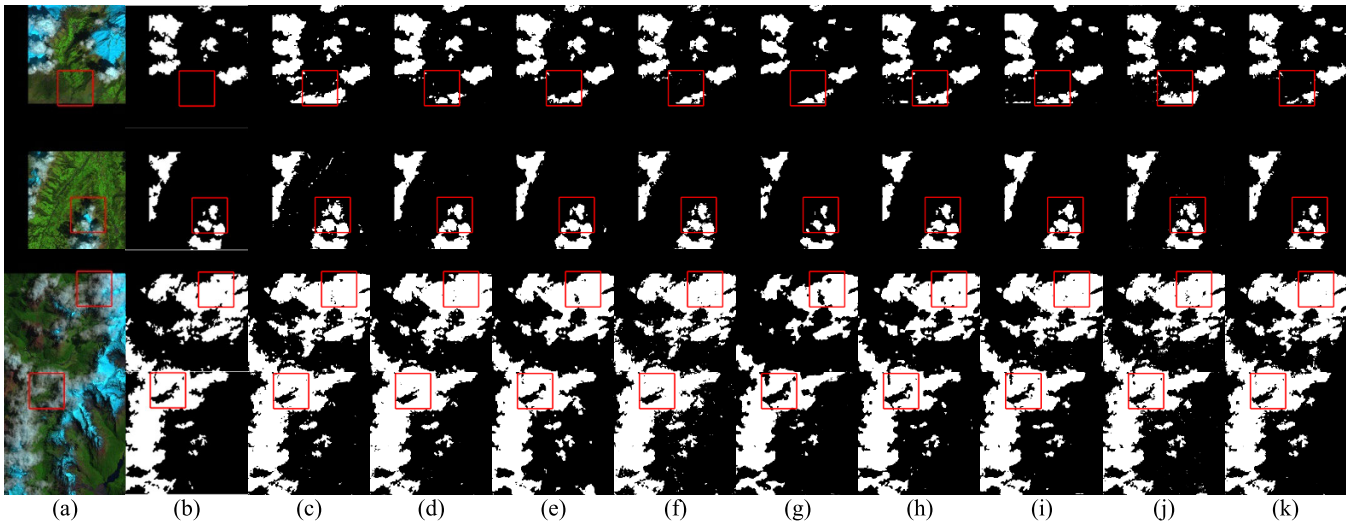


Fig. 14. Cloud masks of different methods for the snow/ice scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

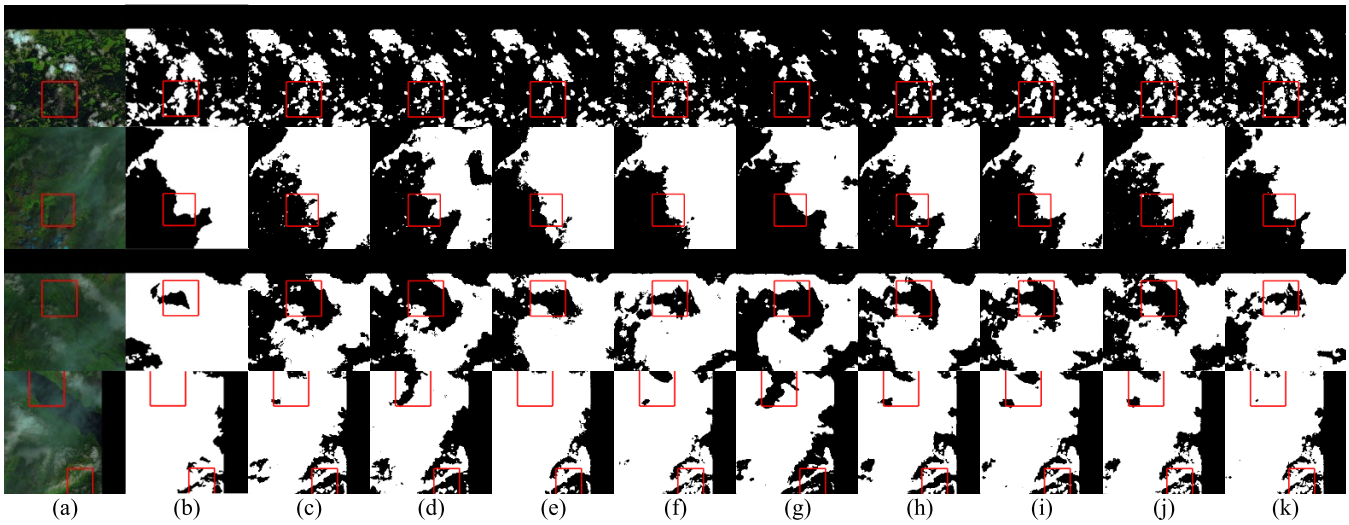


Fig. 15. Cloud masks of different methods for the wetland scenes: (a) false-color image, (b) reference, (c) Cloud-Net, (d) CloudX-Net, (e) AdaptSegNet, (f) RS-Net, (g) DC-Net, (h) MAFA-Net, (i) MSFA-Net, (j) RD-UNet, and (k) AMCD-Net.

multilevel information fusion and different attention mechanisms, AMCD-Net effectively identified both thick clouds and thin clouds over various underlying surfaces, with clear gains in recall, to achieve the optimal OA, F1-score, and Jaccard.

AMCD-Net showed large superiority to the other methods in the most challenging scenarios, i.e., barren and snow/ice, on both two datasets. Compared with the second- and third-best methods, i.e., RD-UNet and RS-Net, for the barren scenario, it achieved clear gains of 4.13% and 1.17% in terms of Jaccard on the 38-Cloud dataset, with more obvious gains of 6.67% and 7.14% obtained on the SPARCS dataset. For the snow/ice scenario, 0.11%, 1.42% and 4.64%, 2.22% gains in Jaccard were obtained on these two datasets. Relative to the other methods, more prominent improvements were achieved. In addition, AMCD-Net obtained obvious gains for the grass/crop scenario over RD-UNet and RS-Net, and showed certain superiority for the water or wetland scenario on both two datasets, which indicates its strong adaptability to different scenarios.

B. Performance Analysis With Different Cloud Amounts

Cloud amount is an important factor for optical remote sensing images and determines the difficulty of cloud detection. For further comparison of different methods, cloud detection performance was individually evaluated in different cases of cloud amount on both the 38-Cloud dataset and the SPARCS dataset, with detailed results provided in Figs. 17 and 18. For convenience, according to the cloud coverage, images in each dataset were divided into three categories, i.e., low (<35%), medium (35%–65%), and high (>65%), with the two most indicative comprehensive metrics, i.e., F1-score and Jaccard, utilized for accuracy evaluation. As illustrated, AMCD-Net achieved the best performance in most cases on both two datasets, which further demonstrates its robustness and generality.

It can be seen that all methods showed underperformance in the case of low cloud amount on the 38-Cloud dataset and in the case of medium cloud amount on the SPARCS dataset, which may be caused by the complex cloud

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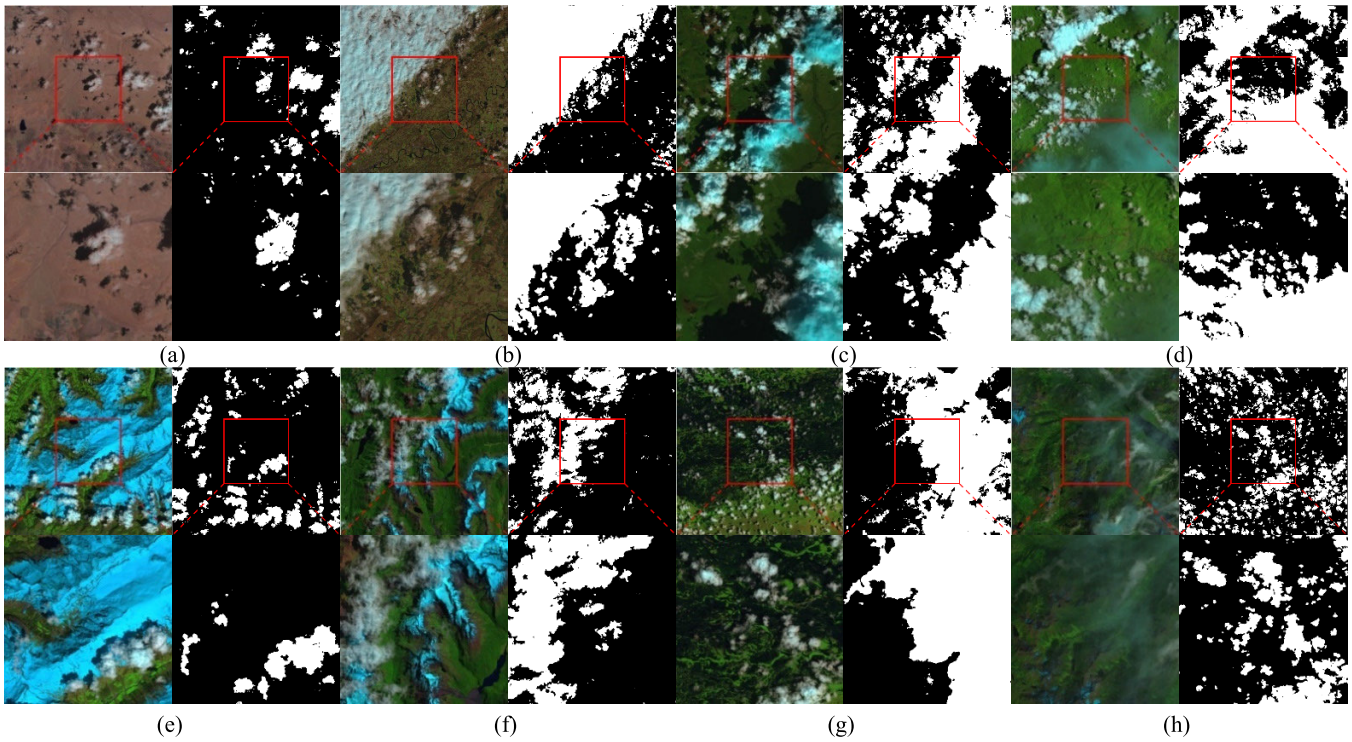


Fig. 16. Typical whole scenes and the corresponding cloud masks obtained by AMCD-Net over various underlying surfaces. (a) Scene ID: LC81400372014158LGN00, barren. (b) Scene ID: LC80380362013144LGN00, barren. (c) Scene ID: LC81220242014240LGN00, grass/crop. (d) Scene ID: LC81140622013277LGN00, grass/crop. (e) Scene ID: LC81950282013156LGN00, snow/ice. (f) Scene ID: LC80760912014013LGN00, snow/ice. (g) Scene ID: LC80470252014226LGN00, wetland. (h) Scene ID: LC82260732013357LGN00, wetland.

TABLE III

QUANTITATIVE EVALUATIONS FOR EACH KIND OF UNDERLYING SURFACES ON THE 38-CLOUD DATASET (%)

Type	Metrics	Cloud-Net	CloudX-Net	Adapt SegNet	RS-Net	DC-Net	MAFA-Net	MSFA-Net	RD-UNet	AMCD-Net
Barren	Precision	77.13	**88.34**	76.41	85.59	77.19	81.49	76.07	81.03	84.01
	Recall	88.20	82.69	82.47	86.58	61.96	79.83	79.76	86.79	**89.56**
	OA	97.77	97.54	97.37	97.91	96.77	97.46	97.45	98.01	**98.02**
	F1-score	80.43	84.75	78.52	85.57	66.19	80.02	76.31	82.49	**86.24**
	Jaccard	70.52	74.25	66.21	75.35	53.47	67.93	64.49	72.39	**76.52**
Grass/crop	Precision	92.37	90.85	92.63	92.26	86.32	88.49	90.79	**95.97**	94.02
	Recall	78.81	83.87	80.92	83.97	71.27	80.40	80.91	84.52	**87.09**
	OA	97.35	96.88	97.18	97.55	95.49	97.25	97.36	**97.61**	97.42
	F1-score	79.45	85.07	81.79	85.30	75.41	81.81	82.79	86.35	**88.10**
	Jaccard	75.52	79.08	77.04	79.90	69.52	77.26	77.95	81.00	**82.09**
Snow/ice	Precision	94.98	92.22	94.69	94.64	93.28	94.81	94.49	**95.42**	94.14
	Recall	87.90	86.44	88.35	88.93	85.94	87.44	87.63	89.56	**90.84**
	OA	94.55	94.02	95.03	94.84	94.17	94.81	94.80	95.27	**95.40**
	F1-score	90.84	88.22	91.12	91.26	89.08	90.72	90.59	92.11	**92.17**
	Jaccard	84.08	81.16	84.59	84.73	81.68	83.95	83.83	86.04	**86.15**
Water	Precision	99.22	**99.54**	95.48	99.44	97.39	99.29	99.01	98.89	99.01
	Recall	67.86	70.24	74.62	78.02	68.93	69.97	69.51	77.40	**78.39**
	OA	89.07	89.92	90.22	92.50	88.89	89.77	89.56	92.14	**92.51**
	F1-score	80.56	82.11	83.64	87.43	80.54	81.96	81.61	86.78	**87.48**
	Jaccard	67.50	69.98	72.02	77.67	67.45	69.59	69.02	76.71	**77.78**

distribution and visually invisible thin clouds. Nevertheless, AMCD-Net achieved acceptable F1-scores of nearly 79% and 77% in these two cases, owing to its strong discriminability. In addition, most methods performed well in the case of high cloud amounts, with considerable precisions. On the other hand, it can be observed that DC-Net performed badly in various cloud amount cases, with low-level precisions obtained. CloudX-Net, RS-Net, and RD-UNet showed obvious advantages in the case of low cloud amount, while MAFA-Net, MSFA-Net, and RD-UNet showed large potential

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TABLE IV
QUANTITATIVE EVALUATIONS FOR EACH KIND OF UNDERLYING SURFACES ON THE SPARCS DATASET (%)

Type	Metrics	Cloud-Net	CloudX-Net	Adapt SegNet	RS-Net	DC-Net	MAFA-Net	MSFA-Net	RD-UNet	AMCD-Net
Barren	Precision	89.96	92.96	90.54	91.02	**95.11**	89.90	94.22	90.60	90.38
	Recall	56.29	57.85	58.22	63.91	49.63	63.56	66.95	64.09	**72.69**
	OA	85.88	87.65	87.13	88.19	86.92	88.33	89.91	88.93	**90.24**
	F1-score	61.71	64.26	64.67	69.04	58.74	67.62	72.50	68.62	**76.47**
	Jaccard	51.14	54.92	54.37	59.15	48.31	58.47	63.09	59.62	**66.29**
Grass/crop	Precision	90.51	91.64	91.28	92.21	**98.57**	89.52	90.72	90.85	91.92
	Recall	91.60	90.06	88.86	92.50	63.67	91.45	91.24	92.20	**93.44**
	OA	94.96	94.79	94.21	95.81	89.68	94.47	94.92	95.15	**95.92**
	F1-score	90.45	90.25	89.41	91.95	75.67	90.01	90.65	91.06	**92.33**
	Jaccard	83.02	82.67	81.35	85.41	62.91	82.15	83.20	83.96	**86.01**
Snow/ice	Precision	80.61	79.88	83.56	84.18	**95.42**	84.42	83.61	81.63	86.47
	Recall	96.31	95.46	94.87	96.64	73.03	93.02	92.66	96.61	**96.71**
	OA	96.48	96.25	96.87	97.18	96.01	96.84	96.66	96.70	**97.59**
	F1-score	87.76	86.96	88.82	89.96	82.71	88.51	87.89	88.46	**91.30**
	Jaccard	78.19	76.93	79.91	81.77	70.53	79.39	78.41	79.35	**83.99**
Wetland	Precision	89.92	91.22	89.98	91.12	**97.54**	89.76	89.90	90.02	90.05
	Recall	88.90	86.53	86.69	90.30	60.49	87.51	86.76	89.63	**91.84**
	OA	92.91	92.80	92.43	93.78	87.70	92.68	92.55	93.13	**94.04**
	F1-score	89.05	88.54	87.98	90.47	73.33	88.46	88.03	89.52	**90.76**
	Jaccard	80.33	79.51	78.60	82.64	59.22	79.33	78.69	81.10	**83.10**

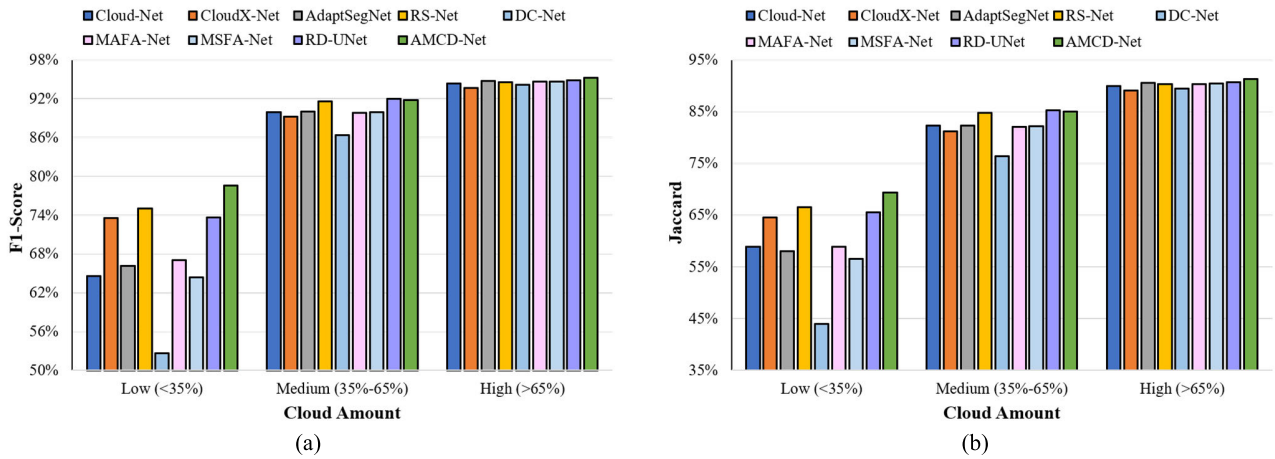


Fig. 17. Performance analysis in cases of different cloud amounts on the 38-Cloud dataset with different metrics: (a) F1-score and (b) Jaccard.

in the case of high cloud amount. In contrast, AMCD-Net exhibited stable performance in all cases to yield competitive accuracy.

Specifically, on the 38-Cloud dataset, AMCD-Net significantly outperformed the other methods in the case of low cloud amount to achieve the highest F1-score and Jaccard of 78.56% and 69.43%, with noticeable improvements over CloudX-Net, RS-Net, and RDU-Net. It performed best in the case of high cloud amount as well to obtain the optimal F1-score and Jaccard, while, in the case of medium cloud amount, AMCD-Net achieved comparable precisions to RDU-Net and RS-Net and showed large advantages to the other methods. Moreover, on the SPARCS dataset, AMCD-Net outperformed the other methods in all cases, with obvious gains in F1-score and Jaccard. Overall, AMCD-Net had a strong adaptability to different cloud amounts.

C. Ablation Study

In the proposed AMCD-Net, four key components were included, i.e., the MAC module, the TCRA block, the GEA, and the joint loss function, to improve cloud detection performance in complex scenarios. To explore the effectiveness of each component, an ablation study was implemented on both the 38-Cloud dataset and the SPARCS dataset, with detailed results reported in Tables V and VI. In addition, cloud heat maps generated by networks with different configurations were visualized to indicate the function of each component, as shown in Fig. 19.

1) MAC Ablation: The MAC module plays an important role in feature encoding, by taking advantage of asymmetric convolutions to learn expressive and direction-aware features for clouds with complex shapes and making full use of multilevel features through multipath cross-layer information

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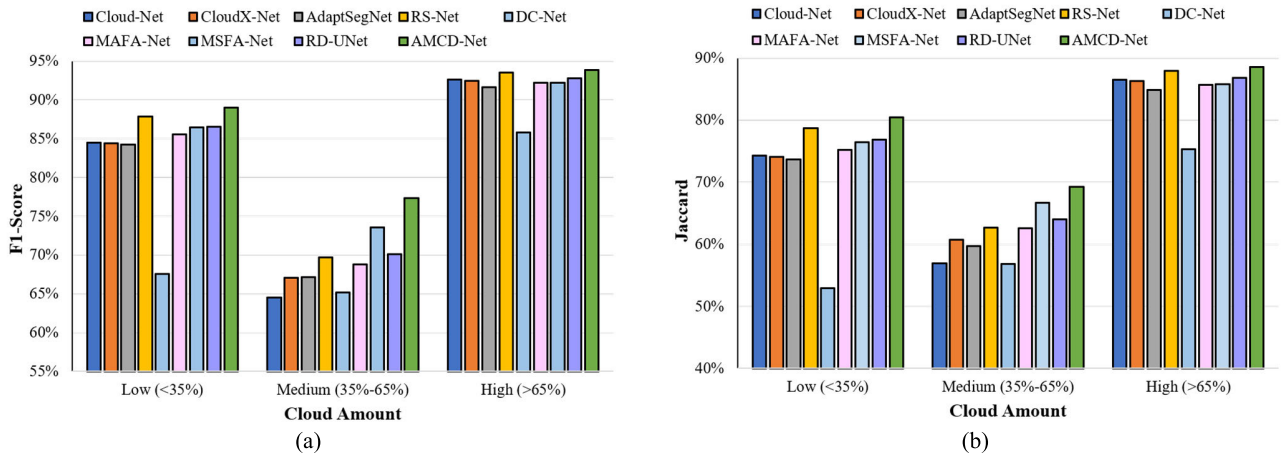


Fig. 18. Performance analysis in cases of different cloud amounts on the SPARCS dataset with different metrics: (a) F1-score and (b) Jaccard.

TABLE V
ABLATION STUDY WITH DIFFERENT COMPONENT COMBINATIONS ON THE 38-CLOUD DATASET (%)

Backbone	MAC	TCRA	GEA	Loss	Precision	Recall	OA	F1-score	Jaccard
√				BCE	90.22	86.92	96.21	87.24	79.87
√	√			BCE	90.90	87.10	96.31	88.01	80.68
√	√	√		BCE	91.41	88.36	**96.56**	88.36	81.44
√	√	√	√	BCE	90.80	88.69	96.55	88.76	81.66
√	√	√	√	BCE+SSIM	**91.82**	**88.78**	96.51	**89.23**	**82.10**

TABLE VI
ABLATION STUDY WITH DIFFERENT COMPONENT COMBINATIONS ON THE SPARCS DATASET (%)

Backbone	MAC	TCRA	GEA	Loss	Precision	Recall	OA	F1-score	Jaccard
√				BCE	**91.13**	83.41	93.13	84.39	76.46
√	√			BCE	90.93	85.05	93.42	85.49	77.68
√	√	√		BCE	90.68	85.86	93.51	86.01	78.00
√	√	√	√	BCE	90.45	86.52	93.75	86.41	78.61
√	√	√	√	BCE+SSIM	90.75	**86.86**	**93.90**	**87.02**	**79.17**

transmission, which effectively improves the discriminability of the model and substantially contributes to thin cloud detection and boundary localization in addition to balancing the integrity and details of clouds. As observed in Tables V and VI, it brings about apparent improvements in most indicators on both two datasets. Specifically, with the help of the MAC module, the F1-scores were improved by 0.77% and 1.1%, with Jaccard indices improved by 0.81% and 1.22%. In addition, from the first row of Fig. 19, it can be found that clouds hidden in the complex background were more completely detected by using the MAC module, with easily missed cirrus clouds and thin clouds effectively identified. As a result, more accurate cloud masks with refined details and boundaries can be obtained.

2) TCRA Ablation: The TCRA block has a crucial effect on feature decoding by fully exploiting regional differences to refine cloud prediction and using transposed convolution instead of linear interpolation for upsampling to reduce detail loss, which is conductive to the accurate restoration of spatial details of clouds and thin cloud identification. As shown in the tables, by adding the TCRA block, obvious gains were obtained in most indicators on both two datasets. The F1-scores were improved by 0.35% and 0.52%, with Jaccard indices improved by 0.76% and 0.32%. Furthermore, as described in the second row of Fig. 19, noncloud regions were correctly judged and thin clouds were more effectively identified with the TCRA block to yield more accurate cloud predictions.

3) GEA Ablation: The GEA has a large impact on information transmission between the encoder and the decoder, which flexibly mines structural features with deformable convolution and emphasizes important information through implicit learning. As a result, it is favorable to finer morphology prediction and boundary localization. From Tables V and VI, clear increments were found in most indicators, with F1-scores improved by 0.4% and 0.4% and Jaccard indices improved by 0.22% and 0.61%, respectively. From the third row of Fig. 19, it can be seen that the irregular shape and

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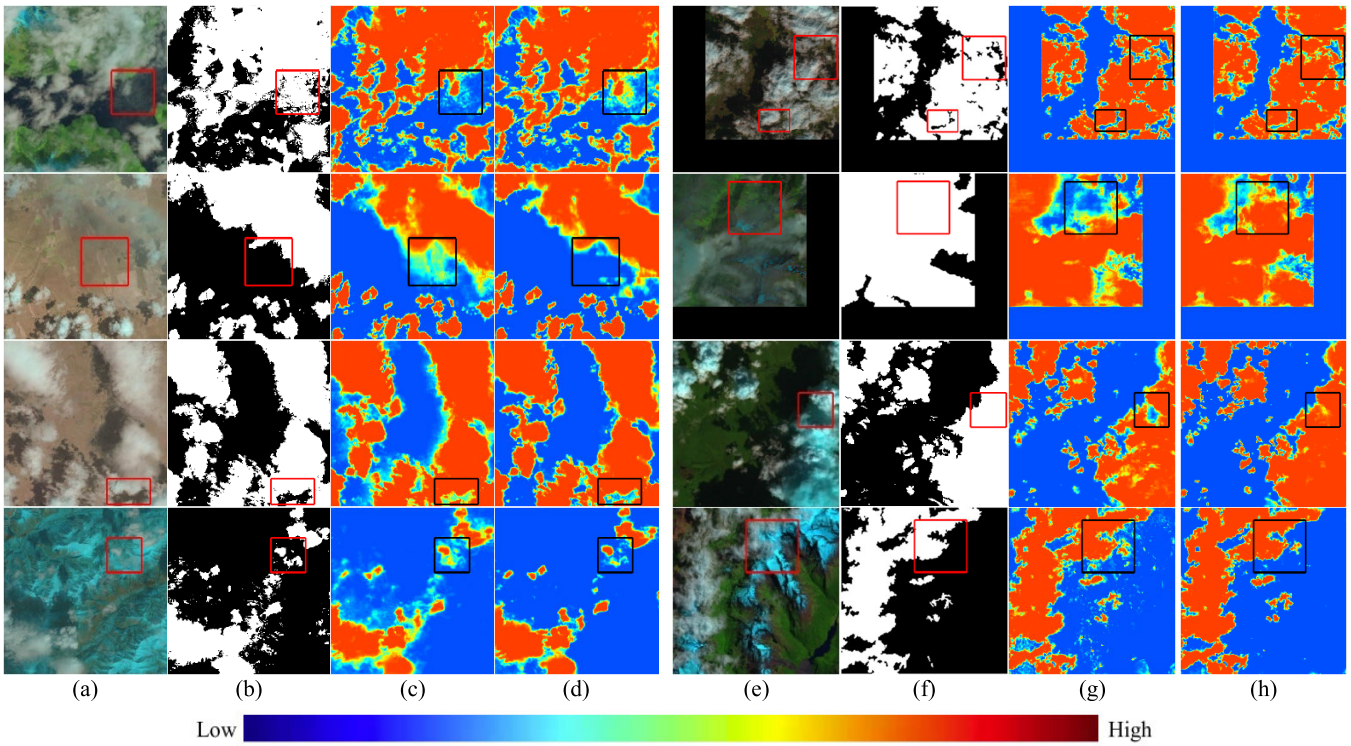


Fig. 19. Visual display of cloud heat maps on the 38-Cloud dataset and the SPARCS dataset: (a) and (e) original images, (b) and (f) references, (c) and (g) (first row) cloud heat maps without MAC, (d) and (h) (first row) cloud heat maps with MAC, (c) and (g) (second row) cloud heat maps without TCRA, (d) and (h) (second row) cloud heat maps with TCRA, (c) and (g) (third row) cloud heat maps without GEA, (d) and (h) (third row) cloud heat maps with GEA, (c) and (f) (fourth row) cloud heat maps with BCE loss, (d) and (h) (fourth row) cloud heat maps with joint loss.

boundary of clouds can be more accurately identified by using GEA.

4) Joint Loss Function Ablation: The joint loss function has a direct influence on the training of the network, which integrates the BCE loss and the SSIM loss to simultaneously promote category and morphology discriminant mining. It is beneficial for improving cloud prediction performance and localizing boundaries. As shown in the tables, evident improvements were obtained in most indicators on both two datasets, with the F1-scores improved by 0.47% and 0.61% and Jaccard indices improved by 0.44% and 0.56%. Moreover, as indicated in the fourth row of Fig. 19, by using the joint loss function, both thick clouds and thin clouds were effectively identified, with detailed structure and accurate boundaries obtained.

By integrating all these components into a unified network, i.e., AMCD-Net, the best performance can be achieved on both two datasets, which clearly demonstrates the effectiveness of these components.

D. Threshold Analysis

The output of AMCD-Net is a cloud probability map, with values ranging from 0 to 1. Therefore, a threshold is required to generate the final cloud mask. Pixels with larger values than threshold are partitioned into clouds, with the other pixels partitioned as background. Generally, the threshold is a tradeoff between precision and recall. A small threshold tends to guarantee few omissions, with a large threshold guaranteeing few commissions.

To find an optimal threshold, a grid-searching strategy was utilized. The threshold changed in the range of [0.05, 0.9], with a step of 0.05. Three comprehensive metrics, i.e., OA, F1-score, and Jaccard, were utilized to indicate the performance change trend of AMCD-Net with threshold variations on both two datasets, as shown in Fig. 20. It can be found that the threshold has a large impact on the final accuracy. However, the performance of AMCD-Net is relatively stable and slightly fluctuates in the range of [0.2, 0.5], with an optimal threshold of 0.3 for the 38-Cloud dataset and an optimal threshold of 0.35 for the SPARCS dataset, which gives good references for real applications and further increases its practicability.

E. Complexity Analysis

For further comparison, the complexity of different methods was analyzed in this section. Herein, the image patch was taken as the basis for complexity analysis, which is the basic processing unit of networks. Two common metrics were utilized, i.e., flop and parameter amount, with detailed results listed in Table VII.

As can be seen, DC-Net and RS-Net have relatively low parameter amounts. Cloud-Net has a much larger parameter amount and a comparable flop, while CloudX-Net has the lowest flop. In contrast, MSFA-Net has a larger parameter amount, and AdaptSegNet, MAFA-Net, and RD-UNet have the largest parameter amounts and flops, with much larger complexity than the other methods. Comparatively speaking, the parameter amount of AMCD-Net increases to some degree

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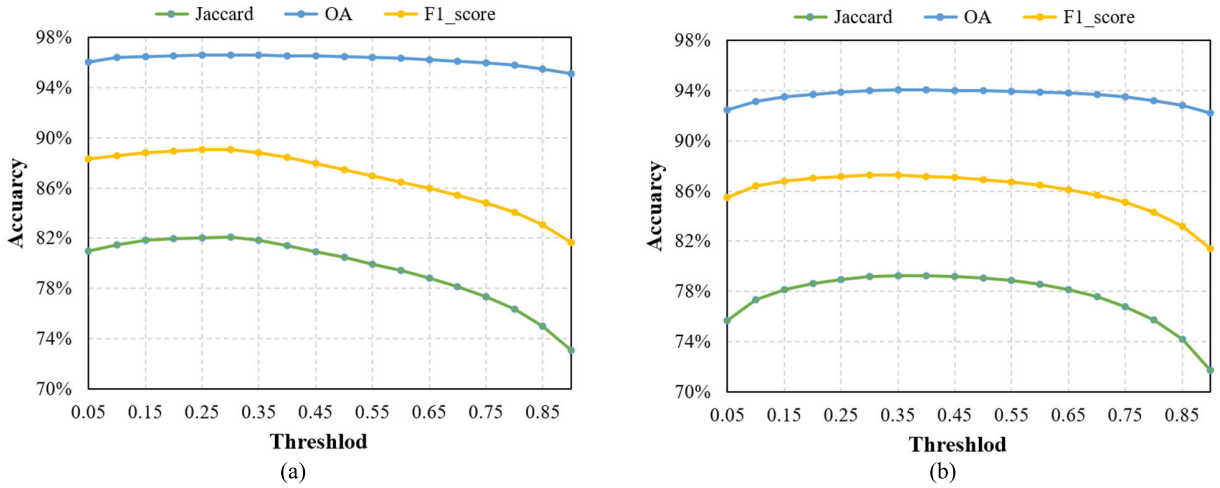


Fig. 20. Threshold analysis for cloud mask generation: (a) 38-Cloud dataset and (b) SPARCS dataset.

TABLE VII
COMPLEXITY ANALYSIS OF DIFFERENT METHODS

Methods	Flops (GB)	Parameters (MB)
Cloud-Net	18.72	36.45
CloudX-Net	**12.87**	19.49
AdaptSegNet	59.37	29.69
RS-Net	15.73	7.85
DC-Net	23.36	**7.12**
MAFA-Net	73.29	36.67
MSFA-Net	38.42	19.23
RD-UNet	62.53	31.27
AMCD-Net	30.00	15.40

relative to DC-Net and RS-Net, as a result of the MAC module, the TCRA block, and the GEA. However, its parameter amount is the third-lowest, and its flop is comparable to that of DC-Net, which is much lower than the other state-of-the-art methods, i.e., MAFA-Net, MSFA-Net, and RD-UNet. Overall, AMCD-Net has an acceptable complexity while yielding the best cloud detection performance, which indicates its good practicability.

VI. CONCLUSION

In this article, a novel AMCD-Net is proposed for optical satellite imagery to promote better performance for various clouds in complex scenarios. A deep convolutional U-Net was adopted as the backbone to learn abstract and invariant features of clouds. On this basis, an innovative MAC module is constructed in the encoder to learn expressive and direction-aware features for complex clouds with asymmetric convolutions and take full advantage of multilevel features through multipath cross-layer information transmission to improve model discriminability. In addition, a regional attention-based decoder is developed to more precisely recover spatial details of clouds by learning regional differentiated importance and introducing transposed convolutions for upsampling to reduce detail loss. Furthermore, an effective GEA is put forward to refine the low-level encoding features to aid cloud prediction by flexibly mining structural features with deformable convolution and emphasizing salient information through implicit learning. Moreover, an efficacy joint loss function consisting of BCE and SSIM is constructed to promote the network to simultaneously focus on category and morphology discriminant excavation in the training process, which further improves cloud prediction performance. Two public datasets, i.e., the 38-Cloud dataset and the SPARCS dataset, were utilized to testify to the effectiveness of the proposed AMCD-Net, with high OA of 96.51% and 93.90%, F1-score of 89.23% and 87.02%, and Jaccard of 82.10% and 79.17% achieved. The experimental results indicate that AMCD-Net outperformed the other state-of-the-art deep networks and is a competitive tool for cloud detection in practical applications.

However, it still has some room for improvement. By adding global information mining modules to make up the locality of convolutions, model discriminability can be further improved. In addition, by introducing more effective structural feature excavation modules, more accurate cloud masks can be obtained. Furthermore, by extending the network to detect clouds and cloud shadows, its practicability can be further enhanced. All of these will be the focus of our next efforts.

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